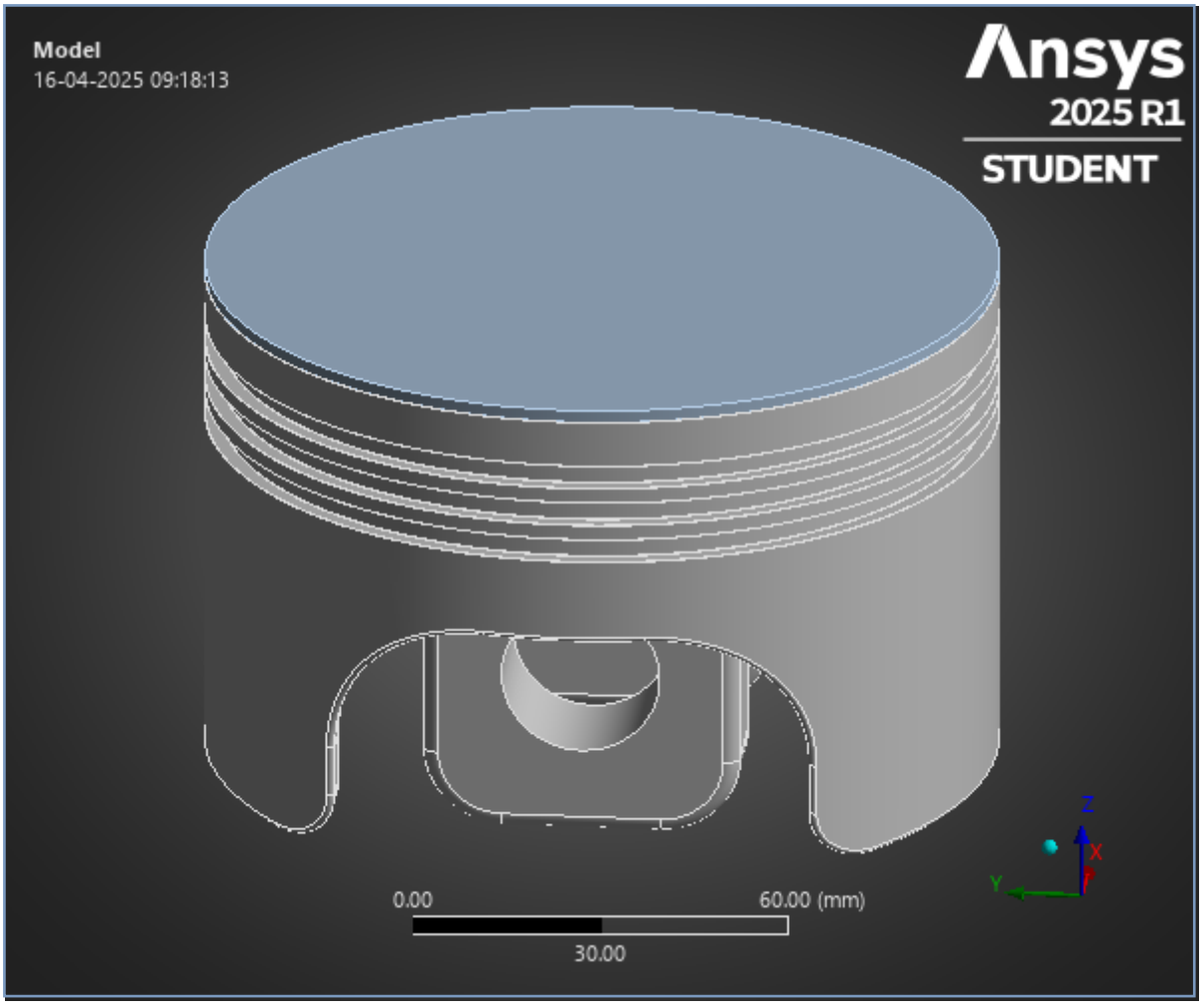
From:
To:
Date: 4/16/2025 3:51:27 AM
Subject:

Project*

First Saved	Wednesday, April 16, 2025
Last Saved	Wednesday, April 16, 2025
Product Version	2025 R1
Save Project Before Solution	No
Save Project After Solution	No



Contents

- Units
- Model (A4)
  - Geometry Imports
    - Geometry Import (A3)
  - Geometry
    - Parts
  - Materials
  - Coordinate Systems
  - Connections
    - Contacts
      - Contact Region
  - Mesh
  - Transient Thermal (A5)
    - Initial Temperature
    - Analysis Settings
    - Loads
    - Solution (A6)
      - Solution Information
        - Result Charts
      - Results
- Material Data
  - Aluminum Alloy
  - TBC coating

Report Not Finalized

Not all objects described below are in a finalized state. As a result, data may be incomplete, obsolete or in error. View first state problem. To finalize this report, edit objects as needed and solve the analyses.

Units

TABLE 1

Unit System	Metric (mm, kg, N, s, mV, mA) Degrees rad/s Celsius
Angle	Degrees
Rotational Velocity	rad/s
Temperature	Celsius

Model (A4)

TABLE 2
Model (A4) > Geometry Imports

Object Name	*Geometry Imports*
State	Solved

TABLE 3
Model (A4) > Geometry Imports > Geometry Import (A3)

Object Name	*Geometry Import (A3)*
State	Solved
Definition	
Source	D:\HEAT TRANSFER PROECT\transient 2mm_files\dp0\SYS\DM\SYS.dsco
Type	Discovery
Basic Geometry Options	
Solid Bodies	Yes
Surface Bodies	Yes

Line Bodies	Yes
Parameters	Independent
Parameter Key	
Attributes	Yes
Attribute Key	
Named Selections	Yes
Named Selection Key	
Material Properties	No
Advanced Geometry Options	
Use Associativity	Yes
Coordinate Systems	Yes
Coordinate System Key	
Reader Mode Saves Updated File	No
Use Instances	Yes
Smart CAD Update	Yes
Compare Parts On Update	No
Analysis Type	3-D
Mixed Import Resolution	None
Import Facet Quality	Source
Clean Bodies On Import	No
Stitch Surfaces On Import	None
Decompose Disjoint Geometry	Yes
Enclosure and Symmetry Processing	Yes

Geometry

TABLE 4
Model (A4) > Geometry

Object Name	*Geometry*
State	Fully Defined
Definition	
Source	D:\HEAT TRANSFER PROECT\transient 2mm_files\dp0\SYS\DM\SYS.dsco
Type	Discovery
Length Unit	Meters
Element Control	Program Controlled
Display Style	Body Color
Bounding Box	
Length X	127. mm
Length Y	127. mm
Length Z	84.55 mm
Properties	
Volume	4.6356e+005 mm³
Mass	1.3632 kg
Scale Factor Value	1.
Statistics	
Bodies	2
Active Bodies	2
Nodes	51760
Elements	25638
Mesh Metric	None
Update Options	
Assign Default Material	No
Basic Geometry Options	
Solid Bodies	Yes
Surface Bodies	Yes
Line Bodies	Yes
Parameters	Independent
Parameter Key	
Attributes	Yes
Attribute Key	
Named Selections	Yes

Named Selection Key	
Material Properties	No
Advanced Geometry Options	
Use Associativity	Yes
Coordinate Systems	Yes
Coordinate System Key	
Reader Mode Saves Updated File	No
Use Instances	Yes
Smart CAD Update	Yes
Compare Parts On Update	No
Analysis Type	3-D
Mixed Import Resolution	None
Import Facet Quality	Source
Clean Bodies On Import	No
Stitch Surfaces On Import	None
Decompose Disjoint Geometry	Yes
Enclosure and Symmetry Processing	Yes

TABLE 5
Model (A4) > Geometry > Parts

Object Name	*Piston:1\Body1*	*Piston Assembly v14\Body1*
State	Meshed	
Graphics Properties		
Visible	Yes	
Transparency	1	
Definition		
Suppressed	No	
Stiffness Behavior	Flexible	
Coordinate System	Default Coordinate System	
Reference Temperature	By Environment	
Treatment	None	
Material		
Assignment	Aluminum Alloy	TBC coating
Nonlinear Effects	Yes	
Thermal Strain Effects	Yes	
Bounding Box		
Length X	127. mm	126.9 mm
Length Y	127. mm	126.9 mm
Length Z	82.55 mm	2. mm
Properties		
Volume	4.3827e+005 mm³	25285 mm³
Mass	1.214 kg	0.14918 kg
Centroid X	-8.6651e-004 mm	-2.1106e-007 mm
Centroid Y	-1.4374e-003 mm	-1.5813e-016 mm
Centroid Z	-27.291 mm	1. mm
Moment of Inertia Ip1	2036. kg·mm²	150.11 kg·mm²
Moment of Inertia Ip2	1846.9 kg·mm²	150.11 kg·mm²
Moment of Inertia Ip3	2569.5 kg·mm²	300.13 kg·mm²
Statistics		
Nodes	44357	7403
Elements	24608	1030
Mesh Metric	None	
CAD Attributes		
PartTolerance:	0.00000001	
Color:130.130.130		
Color:175.146.143		

TABLE 6
Model (A4) > Materials

Object Name	*Materials*
State	Fully Defined

Statistics	
Materials	3
Material Assignments	0

Coordinate Systems

TABLE 7
Model (A4) > Coordinate Systems > Coordinate System

Object Name	*Global Coordinate System*
State	Fully Defined
Definition	
Type	Cartesian
Coordinate System ID	0.
Origin	
Origin X	0. mm
Origin Y	0. mm
Origin Z	0. mm
Directional Vectors	
X Axis Data	[1. 0. 0.]
Y Axis Data	[0. 1. 0.]
Z Axis Data	[0. 0. 1.]
Transfer Properties	
Source	
Read Only	No

Connections

TABLE 8
Model (A4) > Connections

Object Name	*Connections*
State	Fully Defined
Auto Detection	
Generate Automatic Connection On Refresh	Yes
Transparency	
Enabled	Yes
Statistics	
Contacts	1
Active Contacts	1
Joints	0
Active Joints	0
Beams	0
Active Beams	0
Bearings	0
Active Bearings	0
Springs	0
Active Springs	0
Body Interactions	0
Active Body Interactions	0

TABLE 9
Model (A4) > Connections > Contacts

Object Name	*Contacts*
State	Fully Defined
Definition	
Connection Type	Contact
Scope	
Scoping Method	Geometry Selection
Geometry	All Bodies
Auto Detection	
Tolerance Type	Slider
Tolerance Slider	0.

Tolerance Value	0.49628 mm
Use Range	No
Face/Face	Yes
Face-Face Angle Tolerance	75. °
Face Overlap Tolerance	Off
Cylindrical Faces	Include
Face/Edge	No
Edge/Edge	No
Priority	Include All
Group By	Bodies
Search Across	Bodies
Statistics	
Connections	1
Active Connections	1

TABLE 10
Model (A4) > Connections > Contacts > Contact Regions

Object Name	*Contact Region*
State	Fully Defined
Scope	
Scoping Method	Geometry Selection
Contact	2 Faces
Target	1 Face
Contact Bodies	Piston:1\Body1
Target Bodies	Piston Assembly v14\Body1
Protected	No
Definition	
Type	Bonded
Scope Mode	Automatic
Behavior	Program Controlled
Trim Contact	Program Controlled
Trim Tolerance	0.49628 mm
Contact APDL Name	
Target APDL Name	
Suppressed	No
Display	
Element Normals	No
Advanced	
Formulation	Program Controlled
Small Sliding	Program Controlled
Detection Method	Program Controlled
Elastic Slip Tolerance	Program Controlled
Thermal Conductance	Program Controlled
Pinball Region	Program Controlled
Geometric Modification	
Contact Geometry Correction	None
Target Geometry Correction	None

Mesh

TABLE 11
Model (A4) > Mesh

Object Name	*Mesh*
State	Solved
Display	
Display Style	Use Geometry Setting
Defaults	
Physics Preference	Mechanical
Element Order	Program Controlled
Element Size	5.0 mm
Sizing	

Use Adaptive Sizing	Yes
Resolution	Default (2)
Mesh Defeaturing	Yes
Defeature Size	Default
Transition	Fast
Span Angle Center	Coarse
Initial Size Seed	Assembly
Bounding Box Diagonal	198.51 mm
Average Surface Area	881.93 mm²
Minimum Edge Length	0.63636 mm
Quality	
Check Mesh Quality	Yes, Errors
Error Limits	Aggressive Mechanical
Target Element Quality	Default (5.e-002)
Smoothing	Medium
Mesh Metric	None
Inflation	
Use Automatic Inflation	None
Inflation Option	Smooth Transition
Transition Ratio	0.272
Maximum Layers	5
Growth Rate	1.2
Inflation Algorithm	Pre
Inflation Element Type	Wedges
View Advanced Options	No
Advanced	
Number of CPUs for Parallel Part Meshing	Program Controlled
Straight Sided Elements	No
Rigid Body Behavior	Dimensionally Reduced
Triangle Surface Mesher	Program Controlled
Topology Checking	Yes
Pinch Tolerance	Please Define
Generate Pinch on Refresh	No
Automatic Methods	
Sheet Body Method	Quad Dominant
Sweepable Body Method	Sweep
Statistics	
Nodes	51760
Elements	25638
Show Detailed Statistics	No

Transient Thermal (A5)

TABLE 12
Model (A4) > Analysis

Object Name	*Transient Thermal (A5)*
State	Solved
Definition	
Physics Type	Thermal
Analysis Type	Transient
Solver Target	Mechanical APDL
Options	
Generate Input Only	No

TABLE 13
Model (A4) > Transient Thermal (A5) > Initial Condition

Object Name	*Initial Temperature*
State	Fully Defined
Definition	
Initial Temperature	Uniform Temperature

Initial Temperature Value	22. °C

TABLE 14
Model (A4) > Transient Thermal (A5) > Analysis Settings

Object Name	*Analysis Settings*
State	Fully Defined
Step Controls	
Number Of Steps	71.
Current Step Number	71.
Step End Time	71.1 s
Auto Time Stepping	Program Controlled
Initial Time Step	1.0014e-002 s
Minimum Time Step	1.0014e-003 s
Maximum Time Step	0.10014 s
Time Integration	On
Solver Controls	
Solver Type	Program Controlled
Radiosity Controls	
Radiosity Solver	Program Controlled
Flux Convergence	1.e-004
Maximum Iteration	1000.
Solver Tolerance	1.e-007 W/mm²
Over Relaxation	0.1
Hemicube Resolution	10.
Nonlinear Controls	
Heat Convergence	Program Controlled
Temperature Convergence	Program Controlled
Line Search	Program Controlled
Nonlinear Formulation	Program Controlled
Advanced	
Contact Split (DMP)	Program Controlled
Output Controls	
Output Selection	None
Calculate Thermal Flux	Yes
Contact Data	Yes
Nodal Forces	No
Volume and Energy	Yes
Euler Angles	Yes
General Miscellaneous	No
Contact Miscellaneous	No
Store Results At	All Time Points
Result File Compression	Program Controlled
Analysis Data Management	
Solver Files Directory	D:\HEAT TRANSFER PROECT\transient 2mm_files\dp0\SYS\MECH\
Future Analysis	None
Scratch Solver Files Directory	
Save MAPDL db	No
Contact Summary	Program Controlled
Delete Unneeded Files	Yes
Nonlinear Solution	Yes
Solver Units	Active System
Solver Unit System	nmm

TABLE 15
Model (A4) > Transient Thermal (A5) > Analysis Settings
Step-Specific "Step Controls"

Step	Step End Time	Initial Time Step	Minimum Time Step	Maximum Time Step
1	1.0014 s	1.0014e-002 s	1.0014e-003 s	0.10014 s
2	2.0028 s			
3	3.0042 s			
4	4.0056 s			
5	5.007 s			

6	6.0085 s	1.0015e-002 s	1.0015e-003 s	0.10015 s
7	7.0099 s	1.0014e-002 s	1.0014e-003 s	0.10014 s
8	8.0113 s			
9	9.0127 s			
10	10.014 s			
11	11.015 s			
12	12.017 s			
13	13.018 s			
14	14.02 s			
15	15.021 s			
16	16.023 s			
17	17.024 s			
18	18.025 s	1.0015e-002 s	1.0015e-003 s	0.10015 s
19	19.027 s	1.0014e-002 s	1.0014e-003 s	0.10014 s
20	20.028 s			
21	21.03 s			
22	22.031 s			
23	23.032 s			
24	24.034 s			
25	25.035 s			
26	26.037 s			
27	27.038 s			
28	28.039 s			
29	29.041 s			
30	30.042 s	1.0015e-002 s	1.0015e-003 s	0.10015 s
31	31.044 s	1.0014e-002 s	1.0014e-003 s	0.10014 s
32	32.045 s			
33	33.047 s			
34	34.048 s			
35	35.049 s			
36	36.051 s			
37	37.052 s			
38	38.053 s			
39	39.055 s			
40	40.056 s			
41	41.058 s			
42	42.059 s	1.0015e-002 s	1.0015e-003 s	0.10015 s
43	43.061 s	1.0014e-002 s	1.0014e-003 s	0.10014 s
44	44.062 s			
45	45.063 s			
46	46.065 s			
47	47.066 s			
48	48.068 s			
49	49.069 s			
50	50.07 s			
51	51.072 s			
52	52.073 s			
53	53.075 s			
54	54.076 s	1.0015e-002 s	1.0015e-003 s	0.10015 s
55	55.078 s	1.0014e-002 s	1.0014e-003 s	0.10014 s
56	56.079 s			
57	57.08 s			
58	58.082 s			
59	59.083 s			
60	60.084 s			
61	61.086 s			
62	62.087 s			
63	63.089 s			
64	64.09 s			
65	65.091 s			
66	66.093 s	1.0015e-002 s	1.0015e-003 s	0.10015 s

67	67.094 s	1.0014e-002 s	1.0014e-003 s	0.10014 s
68	68.096 s			
69	69.097 s			
70	70.099 s			
71	71.1 s			

TABLE 16
Model (A4) > Transient Thermal (A5) > Loads

Object Name	*Temperature*	*Convection*
State	Fully Defined	
Scope		
Scoping Method	Geometry Selection	
Geometry	1 Body	
Apply To	Exterior Faces Only	
Definition		
Type	Temperature	Convection
Magnitude	Tabular Data	
Suppressed	No	
Film Coefficient		Tabular Data
Coefficient Type		Difference of Surface and Bulk Temp
Ambient Temperature		22. °C (step applied)
Convection Matrix		Program Controlled
Element APDL Name		
Edit Data For		Film Coefficient
Tabular Data		
Independent Variable	Time	Temperature
Graph Controls		
X-Axis		

FIGURE 1
Model (A4) > Transient Thermal (A5) > Temperature

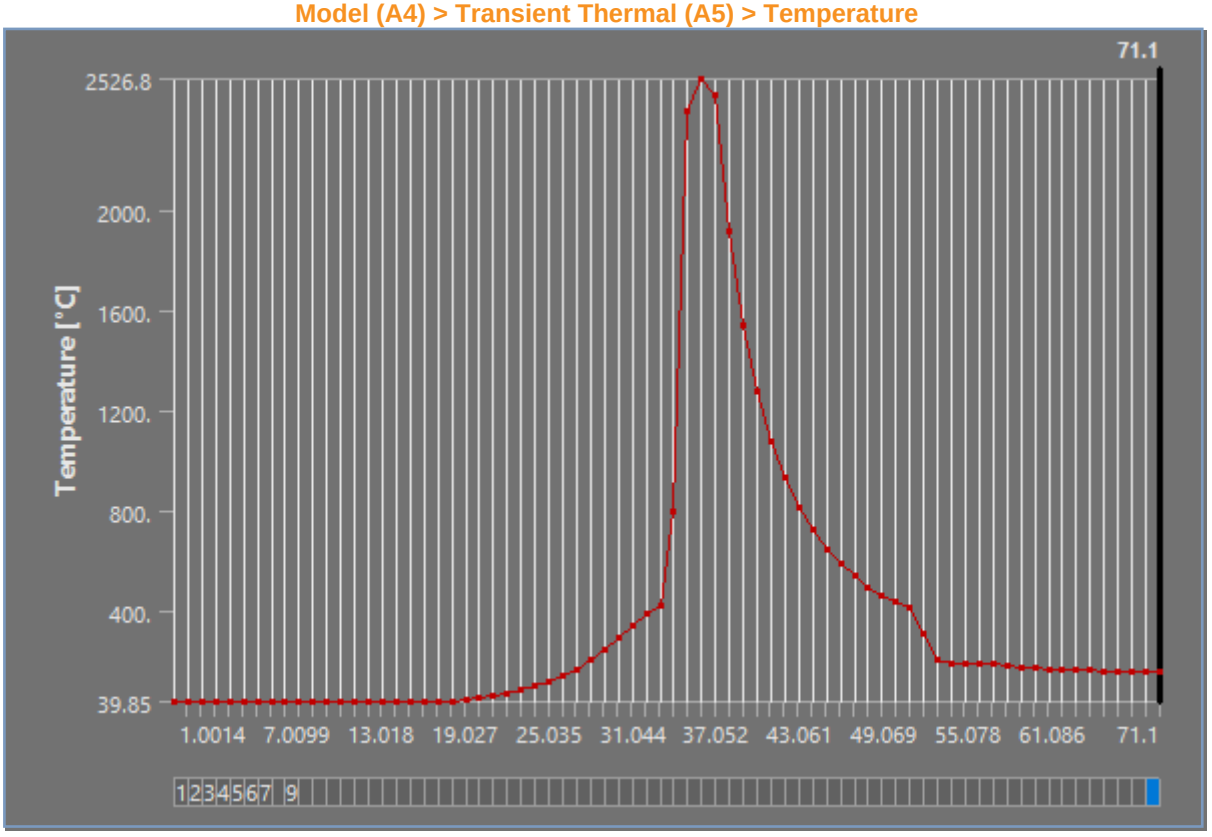


TABLE 17
Model (A4) > Transient Thermal (A5) > Temperature

Steps	Time [s]	Temperature [°C]
1	0.	39.85
	1.0014	
2	2.0028	

3	3.0042	
4	4.0056	
5	5.007	
6	6.0085	
7	7.0099	
8	8.0113	
9	9.0127	
10	10.014	
11	11.015	
12	12.017	
13	13.018	
14	14.02	
15	15.021	
16	16.023	
17	17.024	
18	18.025	39.899
19	19.027	41.062
20	20.028	43.803
21	21.03	48.196
22	22.031	54.358
23	23.032	62.462
24	24.034	72.744
25	25.035	85.52
26	26.037	101.2
27	27.038	120.31
28	28.039	143.53
29	29.041	171.67
30	30.042	205.63
31	31.044	246.19
32	32.045	293.32
33	33.047	344.6
34	34.048	392.55
35	35.049	423.26
36	36.051	799.
37	37.052	2397.4
38	38.053	2526.8
39	39.055	2463.5
40	40.056	1918.6
41	41.058	1546.3
42	42.059	1280.3
43	43.061	1083.4
44	44.062	933.49
45	45.063	816.81
46	46.065	724.31
47	47.066	649.92
48	48.068	589.46
49	49.069	539.92
50	50.07	499.14
51	51.072	465.53
52	52.073	437.91
53	53.075	415.37
54	54.076	313.04
55	55.078	210.56
56	56.079	190.14
57	57.08	
58	58.082	189.85
59	59.083	
60	60.084	184.81
61	61.086	179.47
62	62.087	174.87
63	63.089	171.04

64	64.09	167.98
65	65.091	165.6
66	66.093	163.81
67	67.094	162.5
68	68.096	161.58
69	69.097	160.95
70	70.099	160.53
71	71.1	160.26

FIGURE 2
Model (A4) > Transient Thermal (A5) > Convection

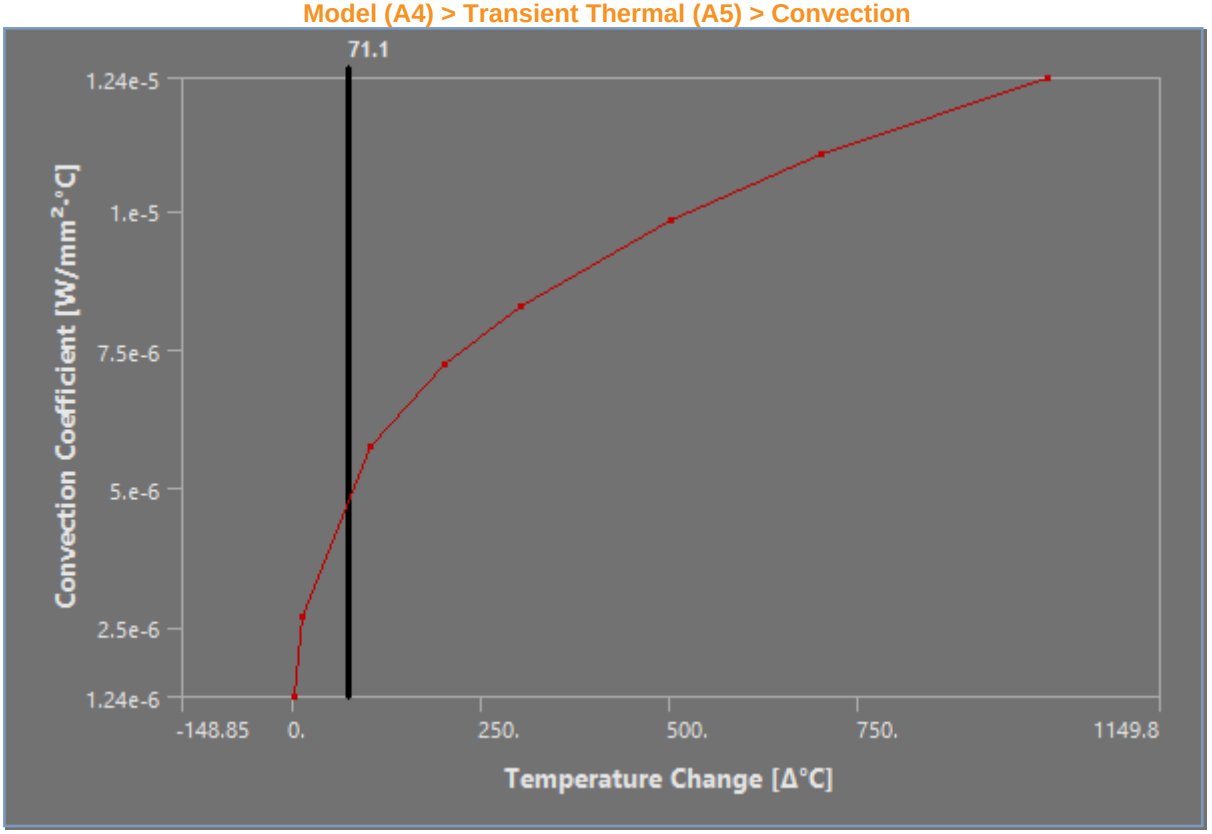


TABLE 18
Model (A4) > Transient Thermal (A5) > Convection

Temperature Change [?°C]	Convection Coefficient [W/mm²·°C]
1.	1.24e-006
10.	2.67e-006
100.	5.76e-006
200.	7.25e-006
300.	8.3e-006
500.	9.84e-006
700.	1.101e-005
1000.	1.24e-005

Solution (A6)

TABLE 19
Model (A4) > Transient Thermal (A5) > Solution

Object Name	*Solution (A6)*
State	Obsolete
Adaptive Mesh Refinement	
Max Refinement Loops	1.
Refinement Depth	2.
Information	
Status	Post-processing Required
MAPDL Elapsed Time	14 m 1 s
MAPDL Memory Used	620. MB

MAPDL Result File Size	6.9097 GB
Post Processing	
Beam Section Results	No

TABLE 20
Model (A4) > Transient Thermal (A5) > Solution (A6) > Solution Information

Object Name	*Solution Information*
State	Solved
Solution Information	
Solution Output	Solver Output
Update Interval	2.5 s
Display Points	All
FE Connection Visibility	
Activate Visibility	Yes
Display	All FE Connectors
Draw Connections Attached To	All Nodes
Line Color	Connection Type
Visible on Results	No
Line Thickness	Single
Display Type	Lines

TABLE 21
Model (A4) > Transient Thermal (A5) > Solution (A6) > Solution Information > Result Charts

Object Name	*Temperature - Global Maximum*	*Temperature - Global Minimum*
State	Solved	
Scope		
Scoping Method	Global Maximum	Global Minimum
Definition		
Type	Temperature	
Suppressed	No	
Results		
Minimum	39.876 °C	11.305 °C
Maximum	2552.1 °C	419.92 °C

FIGURE 3
Model (A4) > Transient Thermal (A5) > Solution (A6) > Solution Information > Temperature - Global Maximum

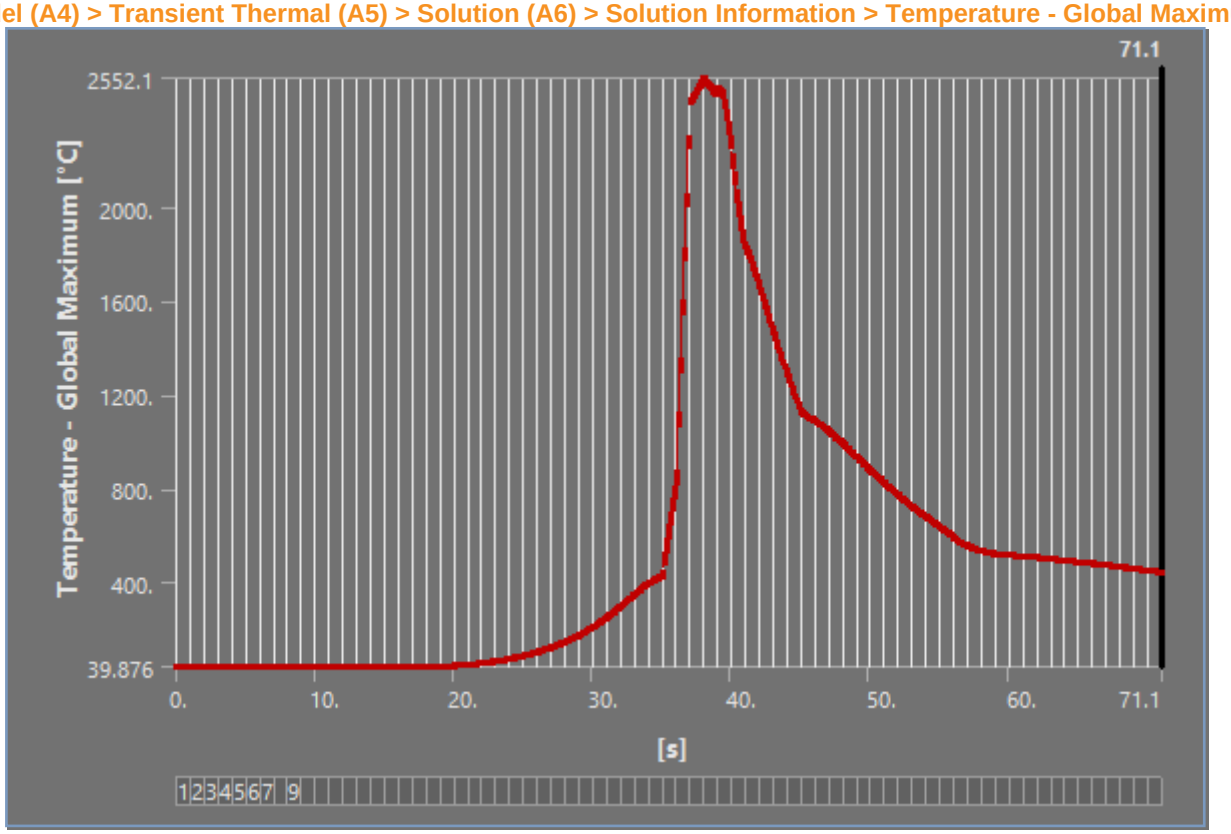


FIGURE 4

Model (A4) > Transient Thermal (A5) > Solution (A6) > Solution Information > Temperature - Global Minimum

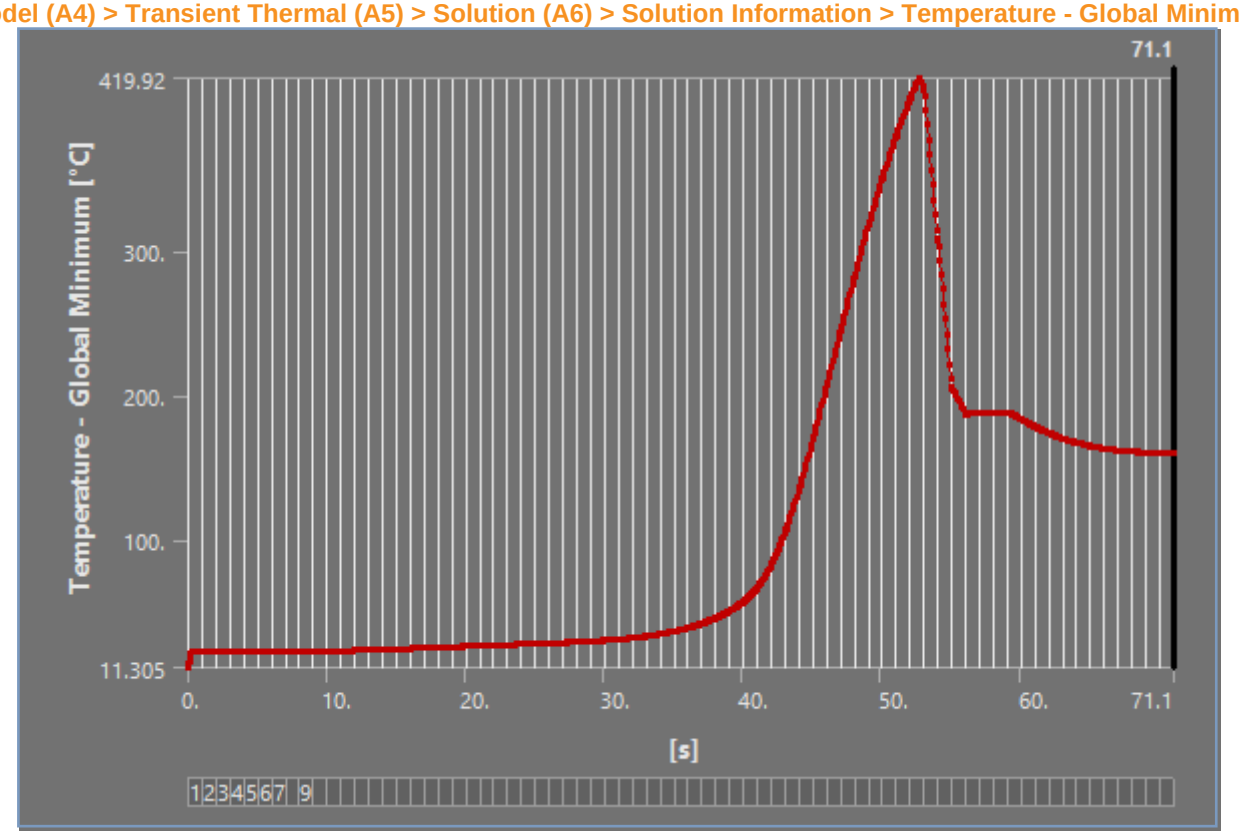


TABLE 22
Model (A4) > Transient Thermal (A5) > Solution (A6) > Results

Object Name	*Temperature*	*Total Heat Flux*
State	Obsolete	Solved
Scope		
Scoping Method	Geometry Selection	Geometry Selection
Geometry	All Bodies	All Bodies
Definition		
Type	Temperature	Total Heat Flux
By	Time	Time
Display Time	First	40. s
Separate Data by Entity	No	No
Calculate Time History	Yes	Yes
Identifier		
Suppressed	No	No
Results		
Minimum	45.511 °C	2.4469e-007 W/mm²
Maximum	2545.7 °C	19.65 W/mm²
Average	828.05 °C	3.3031 W/mm²
Minimum Occurs On	Piston:1\Body1	Piston Assembly v14\Body1
Maximum Occurs On	Piston:1\Body1	Piston:1\Body1
Minimum Value Over Time		
Minimum	11.305 °C	4.7952e-018 W/mm²
Maximum	419.92 °C	1.0225e-006 W/mm²
Maximum Value Over Time		
Minimum	39.876 °C	7.9628e-002 W/mm²
Maximum	2552.1 °C	41.146 W/mm²
Information		
Time	38. s	40. s
Load Step	38	40
Substep	53	30
Iteration Number	621	680
Integration Point Results		
Display Option		Averaged

Average Across Bodies		No

FIGURE 5
Model (A4) > Transient Thermal (A5) > Solution (A6) > Temperature

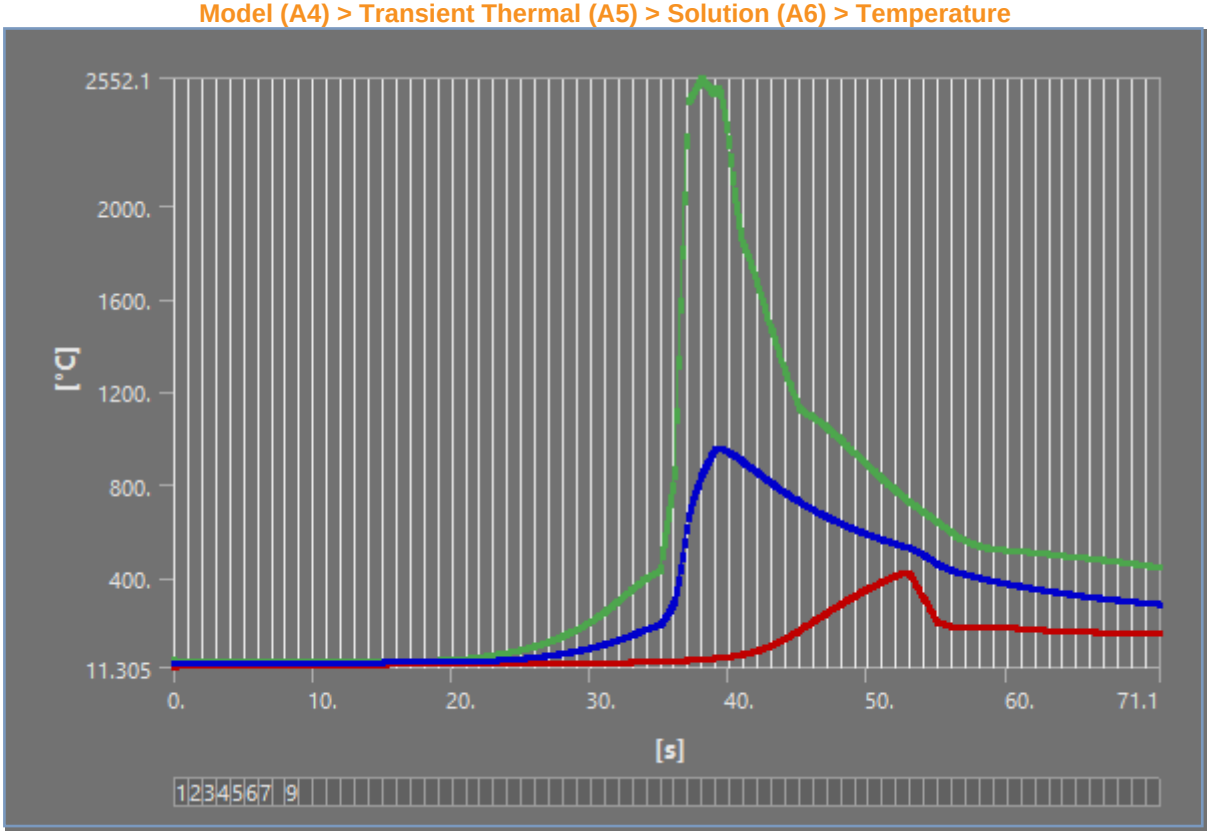


TABLE 23
Model (A4) > Transient Thermal (A5) > Solution (A6) > Temperature

Time [s]	Minimum [°C]	Maximum [°C]	Average [°C]
1.0014e-002	11.305	42.721	24.494
1.9254e-002	11.945	42.	24.449
2.5346e-002	12.361	41.929	24.461
3.1439e-002	12.771	41.852	24.477
4.7761e-002	13.817	41.608	24.524
7.4673e-002	15.418	41.303	24.601
0.11933	17.813	41.003	24.723
0.19327	21.167	40.742	24.906
0.27247		40.576	25.082
0.37261		40.439	25.282
0.47275		40.35	25.463
0.57289		40.286	25.63
0.67303		40.238	25.785
0.77317		40.202	25.93
0.87331		40.173	26.067
0.93735		40.157	26.152
1.0014		40.143	26.234
1.0114		40.139	26.247
1.0214	22.	40.137	26.259
1.0515		40.131	26.297
1.1416		40.116	26.406
1.2417		40.102	26.523
1.3419		40.089	26.635
1.442		40.078	26.743
1.5422		40.068	26.848
1.6423		40.059	26.95
1.7424		40.05	27.049
1.8426		40.043	27.145
1.9427		40.036	27.239

2.0028		40.031	27.295
2.0128			27.304
2.0228		40.03	27.313
2.0529		40.028	27.34
2.143		40.022	27.421
2.2431		40.017	27.509
2.3433		40.011	27.594
2.4434		40.006	27.678
2.5436		40.001	27.76
2.6437		39.997	27.84
2.7438		39.992	27.919
2.844		39.988	27.996
2.9441	22.001	39.984	28.072
3.0042		39.982	28.117
3.0142		39.981	28.124
3.0242			28.132
3.0543		39.98	28.154
3.1444		39.976	28.22
3.2445		39.973	28.291
3.3447	22.002	39.97	28.362
3.4448		39.967	28.431
3.545		39.964	28.499
3.6451	22.003	39.961	28.566
3.7452	22.004	39.958	28.632
3.8454		39.955	28.696
3.9455	22.005	39.953	28.76
4.0056	22.006	39.951	28.798
4.0156			28.804
4.0256			28.81
4.0557		39.95	28.829
4.1458	22.007	39.948	28.885
4.2459	22.009	39.946	28.945
4.3461	22.01	39.944	29.005
4.4462	22.012	39.942	29.064
4.5464	22.014	39.94	29.122
4.6465	22.016	39.938	29.179
4.7466	22.018	39.936	29.236
4.8468	22.02	39.934	29.292
4.9469	22.023	39.933	29.346
5.007	22.025	39.932	29.379
5.017		39.931	29.384
5.027			29.39
5.0571	22.026		29.406
5.1472	22.029	39.929	29.454
5.2474	22.032	39.928	29.507
5.3475	22.036	39.926	29.559
5.4477	22.04	39.925	29.611
5.5478	22.045	39.924	29.661
5.648	22.049	39.922	29.711
5.7481	22.054	39.921	29.761
5.8483	22.059	39.92	29.81
5.9484	22.065	39.919	29.858
6.0085	22.069	39.918	29.887
6.0185			29.892
6.0285	22.07		29.897
6.0586	22.072		29.911
6.1487	22.077	39.917	29.954
6.2488	22.084	39.916	30.
6.349	22.091	39.915	30.047
6.4491	22.099	39.914	30.092
6.5493	22.107	39.913	30.138

6.6494	22.115	39.912	30.182
6.7495	22.124	39.911	30.227
6.8497	22.133	39.91	30.27
6.9498	22.142	39.909	30.314
7.0099	22.148		30.34
7.0199	22.149		30.344
7.0299	22.15	39.908	30.348
7.06	22.153		30.361
7.1501	22.163	39.907	30.399
7.2502	22.173		30.442
7.3504	22.185	39.906	30.483
7.4505	22.196	39.905	30.525
7.5507	22.208		30.565
7.6508	22.221	39.904	30.606
7.7509	22.234	39.903	30.646
7.8511	22.247		30.686
7.9512	22.261	39.902	30.725
8.0113	22.269		30.749
8.0213	22.271	39.901	30.753
8.0313	22.272		30.757
8.0614	22.277		30.768
8.1515	22.29		30.803
8.2516	22.305	39.9	30.842
8.3518	22.321		30.88
8.4519	22.337	39.899	30.918
8.5521	22.353	39.898	30.955
8.6522	22.37		30.992
8.7523	22.387	39.897	31.029
8.8525	22.405		31.066
8.9526	22.423	39.896	31.102
9.0127	22.434		31.123
9.0227	22.436		31.127
9.0327	22.438		31.131
9.0628	22.443		31.141
9.1529	22.46	39.895	31.174
9.253	22.48		31.209
9.3532	22.499		31.244
9.4533	22.52	39.894	31.279
9.5535	22.54		31.314
9.6536	22.561	39.893	31.348
9.7537	22.583		31.382
9.8539	22.604		31.416
9.954	22.627	39.892	31.449
10.014	22.64		31.47
10.024	22.642		31.473
10.034	22.644		31.476
10.064	22.651		31.486
10.154	22.672	39.891	31.516
10.254	22.695		31.549
10.355	22.719		31.582
10.455	22.743	39.89	31.614
10.555	22.767		31.646
10.655	22.792		31.678
10.755	22.817	39.889	31.71
10.855	22.842		31.742
10.955	22.868		31.773
11.016	22.884	39.888	31.792
11.026	22.886		31.795
11.036	22.889		31.798
11.066	22.897		31.807
11.156	22.92		31.835

11.256	22.947		31.866
11.356	22.974	39.887	31.896
11.456	23.002		31.927
11.556	23.029		31.957
11.656	23.057		31.987
11.757	23.085	39.886	32.017
11.857	23.114		32.046
11.957	23.143		32.076
12.017	23.16		32.093
12.027	23.163		32.096
12.037	23.166		32.099
12.067	23.175	39.885	32.108
12.157	23.201		32.134
12.257	23.231		32.163
12.357	23.26		32.191
12.458	23.291	39.884	32.22
12.558	23.321		32.248
12.658	23.351		32.277
12.758	23.382		32.305
12.858	23.413		32.332
12.958	23.445	39.883	32.36
13.018	23.464		32.377
13.028	23.467		32.379
13.038	23.47		32.382
13.068	23.479		32.39
13.158	23.508		32.415
13.259	23.54		32.442
13.359	23.572	39.882	32.469
13.459	23.604		32.496
13.559	23.637		32.523
13.659	23.67		32.55
13.759	23.703		32.576
13.859	23.736	39.881	32.603
13.96	23.769		32.629
14.02	23.789		32.644
14.03	23.792		32.647
14.04	23.796		32.65
14.07	23.806		32.657
14.16	23.836		32.681
14.26	23.87		32.707
14.36	23.904	39.88	32.732
14.46	23.938		32.758
14.56	23.972		32.783
14.661	24.006		32.808
14.761	24.041		32.833
14.861	24.076		32.858
14.961	24.11	39.879	32.883
15.021	24.131		32.898
15.031	24.135		32.9
15.041	24.138		32.903
15.071	24.149		32.91
15.161	24.18		32.932
15.261	24.216		32.957
15.362	24.251		32.981
15.462	24.286		33.005
15.562	24.322	39.878	33.03
15.662	24.357		33.054
15.762	24.393		33.077
15.862	24.429		33.101
15.962	24.465		33.125
16.022	24.486		33.139

16.033	24.49		33.141
16.043	24.493		33.144
16.073	24.504		33.151
16.163	24.537	39.877	33.172
16.263	24.573		33.195
16.363	24.609		33.218
16.463	24.645		33.241
16.563	24.682		33.264
16.663	24.718		33.287
16.764	24.755		33.31
16.864	24.791	39.876	33.332
16.964	24.828		33.355
17.024	24.85		33.368
17.034	24.854	39.877	33.37
17.044	24.857		33.373
17.074	24.868	39.879	33.38
17.164	24.901	39.884	33.4
17.264	24.938	39.889	33.423
17.364	24.975	39.894	33.446
17.465	25.012	39.898	33.469
17.565	25.049	39.903	33.492
17.665	25.086	39.908	33.515
17.765	25.123	39.913	33.538
17.865	25.16	39.918	33.56
17.965	25.197	39.923	33.583
18.025	25.219	39.926	33.597
18.035	25.223	39.939	33.6
18.045	25.226	39.952	33.604
18.075	25.237	39.99	33.616
18.165	25.271	40.101	33.651
18.265	25.308	40.222	33.693
18.365	25.345	40.342	33.735
18.466	25.382	40.462	33.779
18.566	25.419	40.581	33.823
18.666	25.457	40.699	33.869
18.766	25.494	40.818	33.915
18.866	25.531	40.936	33.962
18.966	25.568	41.054	34.01
19.027	25.591	41.125	34.04
19.037	25.594	41.156	34.047
19.047	25.598	41.185	34.054
19.077	25.609	41.272	34.075
19.167	25.643	41.529	34.142
19.267	25.68	41.812	34.219
19.367	25.717	42.093	34.298
19.467	25.754	42.373	34.379
19.568	25.792	42.653	34.463
19.668	25.829	42.932	34.549
19.768	25.866	43.21	34.636
19.868	25.903	43.489	34.726
19.968	25.94	43.766	34.817
20.028	25.963	43.933	34.872
20.038	25.966	43.98	34.884
20.048	25.97	44.026	34.895
20.078	25.981	44.164	34.93
20.168	26.015	44.571	35.038
20.269	26.052	45.021	35.162
20.369	26.089	45.469	35.289
20.469	26.126	45.916	35.42
20.569	26.163	46.363	35.553
20.669	26.2	46.808	35.69

20.769	26.237	47.253	35.828
20.869	26.274	47.698	35.97
20.97	26.311	48.143	36.114
21.03	26.333	48.409	36.201
21.04	26.337	48.474	36.218
21.05	26.341	48.538	36.236
21.08	26.352	48.729	36.287
21.17	26.385	49.298	36.446
21.27	26.422	49.926	36.627
21.37	26.459	50.553	36.813
21.47	26.495	51.179	37.002
21.57	26.532	51.803	37.196
21.67	26.569	52.427	37.392
21.771	26.606	53.051	37.593
21.871	26.642	53.674	37.797
21.971	26.679	54.296	38.003
22.031	26.701	54.67	38.129
22.041	26.705	54.754	38.152
22.051	26.708	54.838	38.176
22.081	26.719	55.088	38.248
22.171	26.752	55.833	38.467
22.271	26.789	56.658	38.717
22.371	26.825	57.48	38.971
22.472	26.862	58.301	39.231
22.572	26.898	59.121	39.495
22.672	26.935	59.941	39.763
22.772	26.971	60.76	40.036
22.872	27.007	61.578	40.313
22.972	27.044	62.396	40.594
23.032	27.065	62.887	40.763
23.042	27.069	62.993	40.795
23.052	27.073	63.099	40.826
23.082	27.084	63.415	40.922
23.173	27.116	64.359	41.213
23.273	27.153	65.403	41.543
23.373	27.189	66.444	41.879
23.473	27.225	67.485	42.221
23.573	27.261	68.524	42.568
23.673	27.298	69.563	42.921
23.773	27.334	70.601	43.278
23.874	27.37	71.638	43.641
23.974	27.407	72.675	44.008
24.034	27.428	73.297	44.231
24.044	27.432	73.429	44.271
24.054	27.436	73.56	44.312
24.084	27.447	73.952	44.435
24.174	27.479	75.122	44.811
24.274	27.516	76.418	45.236
24.374	27.552	77.711	45.668
24.474	27.589	79.003	46.108
24.575	27.625	80.293	46.553
24.675	27.662	81.583	47.005
24.775	27.698	82.872	47.464
24.875	27.735	84.16	47.927
24.975	27.772	85.448	48.397
25.035	27.794	86.22	48.681
25.045	27.798	86.38	48.733
25.055	27.802	86.541	48.784
25.085	27.813	87.022	48.94
25.175	27.846	88.456	49.417
25.276	27.883	90.044	49.955

25.376	27.921	91.63	50.502
25.476	27.958	93.214	51.057
25.576	27.996	94.797	51.619
25.676	28.034	96.378	52.189
25.776	28.072	97.959	52.766
25.876	28.11	99.539	53.351
25.977	28.148	101.12	53.941
26.037	28.171	102.07	54.298
26.047	28.175	102.26	54.363
26.057	28.179	102.46	54.428
26.087	28.191	103.04	54.624
26.177	28.226	104.79	55.22
26.277	28.265	106.72	55.894
26.377	28.304	108.66	56.578
26.477	28.344	110.58	57.271
26.577	28.384	112.51	57.974
26.677	28.425	114.44	58.685
26.778	28.465	116.37	59.405
26.878	28.507	118.29	60.133
26.978	28.548	120.22	60.869
27.038	28.573	121.36	61.318
27.048	28.577	121.6	61.398
27.058	28.581	121.84	61.479
27.088	28.594	122.55	61.722
27.178	28.632	124.67	62.462
27.278	28.675	127.03	63.298
27.378	28.718	129.37	64.145
27.479	28.762	131.71	65.005
27.579	28.806	134.06	65.875
27.679	28.851	136.4	66.756
27.779	28.897	138.74	67.648
27.879	28.943	141.07	68.554
27.979	28.989	143.41	69.47
28.039	29.017	144.81	70.023
28.049	29.022	145.1	70.123
28.059	29.027	145.39	70.222
28.089	29.041	146.25	70.522
28.18	29.084	148.82	71.434
28.28	29.133	151.67	72.464
28.38	29.182	154.51	73.508
28.48	29.232	157.35	74.566
28.58	29.283	160.19	75.637
28.68	29.335	163.02	76.722
28.78	29.388	165.85	77.824
28.881	29.441	168.68	78.938
28.981	29.496	171.51	80.064
29.041	29.529	173.21	80.744
29.051	29.534	173.56	80.865
29.061	29.54	173.91	80.987
29.091	29.556	174.95	81.355
29.181	29.607	178.05	82.473
29.281	29.664	181.49	83.735
29.381	29.723	184.92	85.015
29.481	29.782	188.34	86.311
29.582	29.843	191.77	87.623
29.682	29.905	195.18	88.957
29.782	29.968	198.6	90.305
29.882	30.032	202.02	91.668
29.982	30.098	205.44	93.044
30.042	30.138	207.49	93.875
30.052	30.144	207.9	94.023

30.062	30.151	208.31	94.172
30.092	30.171	209.55	94.619
30.182	30.232	213.25	95.981
30.283	30.302	217.35	97.516
30.383	30.373	221.45	99.072
30.483	30.446	225.54	100.65
30.583	30.52	229.61	102.25
30.683	30.596	233.7	103.86
30.783	30.674	237.78	105.5
30.883	30.753	241.86	107.15
30.984	30.834	245.94	108.83
31.044	30.883	248.39	109.84
31.054	30.891	248.88	110.01
31.064	30.899	249.36	110.19
31.094	30.924	250.8	110.73
31.184	31.001	255.09	112.35
31.284	31.087	259.86	114.19
31.384	31.176	264.61	116.05
31.484	31.266	269.36	117.93
31.584	31.359	274.11	119.83
31.685	31.454	278.86	121.76
31.785	31.551	283.6	123.7
31.885	31.65	288.34	125.67
31.985	31.752	293.08	127.65
32.045	31.814	295.93	128.85
32.055	31.824	296.45	129.05
32.065	31.835	296.97	129.26
32.095	31.866	298.53	129.88
32.185	31.962	303.19	131.76
32.285	32.071	308.36	133.87
32.386	32.183	313.53	136.
32.486	32.297	318.69	138.16
32.586	32.414	323.85	140.33
32.686	32.534	329.01	142.53
32.786	32.657	334.17	144.75
32.886	32.783	339.32	146.98
32.986	32.912	344.48	149.24
33.047	32.991	347.57	150.6
33.057	33.004	348.05	150.82
33.067	33.017	348.53	151.04
33.097	33.057	349.96	151.71
33.187	33.179	354.28	153.72
33.287	33.318	359.09	155.97
33.387	33.46	363.9	158.24
33.487	33.605	368.7	160.51
33.587	33.755	373.51	162.8
33.687	33.907	378.32	165.11
33.788	34.064	383.13	167.42
33.888	34.225	387.94	169.75
33.988	34.389	392.75	172.09
34.048	34.489	395.64	173.5
34.058	34.506	395.92	173.71
34.068	34.523	396.21	173.92
34.098	34.574	397.09	174.56
34.188	34.729	399.78	176.44
34.288	34.906	402.79	178.52
34.388	35.088	405.83	180.6
34.489	35.274	408.86	182.66
34.589	35.464	411.91	184.72
34.689	35.659	414.96	186.78
34.789	35.859	418.01	188.83

34.889	36.064	421.07	190.87
34.989	36.274	424.13	192.92
35.049	36.402	425.96	194.14
35.059	36.424	430.22	194.83
35.069	36.445	434.36	195.51
35.095	36.501	444.95	197.31
35.142	36.604	463.68	200.71
35.19	36.709	482.57	204.3
35.239	36.818	501.54	208.06
35.288	36.928	520.55	211.97
35.337	37.041	539.6	216.01
35.387	37.155	558.66	220.17
35.436	37.271	577.73	224.44
35.486	37.388	596.81	228.83
35.536	37.508	615.9	233.31
35.586	37.629	634.99	237.89
35.636	37.751	654.09	242.56
35.686	37.875	673.19	247.32
35.736	38.001	692.29	252.16
35.786	38.129	711.39	257.09
35.836	38.258	730.5	262.1
35.886	38.389	749.61	267.18
35.936	38.521	768.71	272.34
35.987	38.655	787.82	277.58
36.019	38.741	799.99	280.94
36.051	38.828	812.15	284.32
36.061	38.855	829.94	287.09
36.071	38.882	847.33	289.84
36.082	38.912	866.33	292.89
36.093	38.943	885.32	295.98
36.104	38.973	904.32	299.13
36.115	39.004	923.34	302.33
36.126	39.035	942.37	305.58
36.138	39.067	961.41	308.87
36.149	39.098	980.47	312.21
36.161	39.13	999.54	315.59
36.172	39.162	1018.6	319.01
36.184	39.194	1037.7	322.47
36.195	39.226	1056.8	325.96
36.207	39.259	1075.9	329.49
36.218	39.291	1095.	333.05
36.23	39.324	1114.1	336.65
36.241	39.356	1133.2	340.28
36.253	39.389	1152.3	343.94
36.264	39.422	1171.4	347.64
36.276	39.456	1190.5	351.36
36.288	39.489	1209.6	355.11
36.299	39.522	1228.7	358.89
36.311	39.556	1247.8	362.7
36.322	39.589	1266.9	366.54
36.334	39.623	1286.	370.41
36.346	39.657	1305.1	374.3
36.357	39.691	1324.2	378.22
36.369	39.725	1343.3	382.17
36.381	39.759	1362.4	386.14
36.393	39.794	1381.5	390.14
36.404	39.828	1400.6	394.17
36.416	39.863	1419.7	398.22
36.428	39.897	1438.8	402.29
36.439	39.932	1458.	406.39
36.451	39.967	1477.1	410.51

36.463	40.002	1496.2	414.66
36.474	40.037	1515.3	418.83
36.486	40.072	1534.4	423.02
36.498	40.108	1553.5	427.24
36.51	40.143	1572.6	431.48
36.521	40.179	1591.7	435.74
36.533	40.214	1610.9	440.02
36.545	40.25	1630.	444.33
36.557	40.286	1649.1	448.66
36.568	40.322	1668.2	453.01
36.58	40.358	1687.3	457.38
36.592	40.394	1706.4	461.77
36.604	40.43	1725.5	466.19
36.615	40.467	1744.7	470.62
36.627	40.503	1763.8	475.08
36.639	40.54	1782.9	479.55
36.651	40.577	1802.	484.05
36.663	40.613	1821.1	488.57
36.674	40.65	1840.2	493.1
36.686	40.687	1859.3	497.66
36.698	40.725	1878.5	502.24
36.71	40.762	1897.6	506.83
36.721	40.799	1916.7	511.45
36.733	40.837	1935.8	516.08
36.745	40.874	1954.9	520.74
36.757	40.912	1974.	525.41
36.769	40.95	1993.2	530.11
36.78	40.988	2012.3	534.82
36.792	41.026	2031.4	539.55
36.804	41.064	2050.5	544.3
36.816	41.102	2069.6	549.06
36.828	41.14	2088.7	553.85
36.839	41.179	2107.9	558.65
36.851	41.217	2127.	563.47
36.863	41.256	2146.1	568.31
36.875	41.295	2165.2	573.17
36.887	41.334	2184.3	578.04
36.899	41.373	2203.4	582.94
36.91	41.412	2222.6	587.85
36.922	41.451	2241.7	592.77
36.934	41.491	2260.8	597.72
36.946	41.53	2279.9	602.68
36.958	41.57	2299.	607.66
36.969	41.609	2318.1	612.66
36.981	41.649	2337.3	617.67
36.993	41.689	2356.4	622.7
37.005	41.729	2375.5	627.75
37.017	41.769	2394.6	632.81
37.029	41.809	2413.7	637.89
37.04	41.85	2432.8	642.99
37.052	41.89	2451.8	648.06
37.062	41.924	2451.2	650.35
37.07	41.953	2451.1	652.29
37.079	41.981	2451.	654.22
37.087	42.011		656.22
37.096	42.042	2451.2	658.24
37.105	42.073	2451.4	660.3
37.114	42.105	2451.7	662.38
37.124	42.137	2452.1	664.49
37.133	42.171	2452.6	666.64
37.143	42.205	2453.2	668.82

37.153	42.24	2453.8	671.04
37.164	42.276	2454.5	673.29
37.174	42.313	2455.2	675.57
37.185	42.351	2456.	677.9
37.196	42.39	2456.8	680.26
37.207	42.43	2457.7	682.67
37.219	42.471	2458.7	685.13
37.231	42.513	2459.7	687.63
37.243	42.557	2460.7	690.17
37.255	42.602	2461.8	692.77
37.268	42.648	2463.	695.43
37.281	42.696	2464.2	698.14
37.295	42.745	2465.5	700.92
37.309	42.796	2466.9	703.75
37.323	42.848	2468.3	706.66
37.338	42.903	2469.8	709.63
37.354	42.959	2471.3	712.68
37.37	43.017	2473.	715.82
37.386	43.078	2474.7	719.03
37.403	43.141	2476.4	722.34
37.421	43.206	2478.3	725.74
37.439	43.274	2480.3	729.24
37.458	43.345	2482.3	732.85
37.478	43.42	2484.5	736.58
37.498	43.497	2486.7	740.44
37.52	43.579	2489.1	744.43
37.542	43.664	2491.6	748.56
37.565	43.753	2494.3	752.85
37.59	43.848	2497.	757.3
37.615	43.947	2500.	761.94
37.642	44.052	2503.1	766.77
37.671	44.163	2506.4	771.81
37.7	44.281	2509.9	777.09
37.731	44.402	2513.4	782.41
37.761	44.524	2517.	787.73
37.792	44.648	2520.7	793.04
37.823	44.774	2524.4	798.36
37.854	44.902	2528.1	803.69
37.886	45.032	2531.9	809.03
37.918	45.165	2535.7	814.39
37.95	45.299	2539.6	819.77
37.982	45.436	2543.5	825.17
38.015	45.576	2547.5	830.6
38.034	45.657	2549.8	833.72
38.053	45.739	2552.1	836.83
38.064	45.782	2551.2	838.2
38.074	45.825	2550.2	839.57
38.088	45.886	2549.	841.51
38.122	46.033	2546.1	846.03
38.156	46.182	2543.3	850.49
38.19	46.336	2540.5	855.
38.226	46.494	2537.7	859.54
38.262	46.657	2534.9	864.08
38.299	46.823	2532.1	868.62
38.336	46.993	2529.3	873.16
38.374	47.167	2526.4	877.71
38.413	47.346	2523.6	882.26
38.452	47.53	2520.7	886.83
38.492	47.719	2517.8	891.41
38.533	47.914	2514.8	896.02
38.574	48.115	2511.8	900.65

38.617	48.322	2508.7	905.32
38.66	48.537	2505.6	910.02
38.705	48.758	2502.4	914.76
38.751	48.988	2499.2	919.54
38.797	49.227	2495.9	924.37
38.845	49.475	2492.5	929.25
38.895	49.734	2489.	934.19
38.946	50.004	2485.8	939.2
38.998	50.286	2485.5	944.27
39.026	50.441	2485.2	947.
39.055	50.597	2484.8	949.71
39.065	50.652	2487.6	949.99
39.075	50.707	2490.1	950.28
39.086	50.77	2492.8	950.59
39.118	50.948	2498.4	951.34
39.151	51.131	2502.3	951.98
39.184	51.318	2504.7	952.52
39.217	51.509	2505.6	952.95
39.25	51.702	2505.1	953.27
39.284	51.899	2503.3	953.48
39.317	52.099	2500.3	953.59
39.351	52.301	2496.	953.6
39.385	52.506	2490.7	953.51
39.419	52.714	2484.4	953.33
39.453	52.925	2477.1	953.06
39.487	53.138	2469.1	952.69
39.521	53.354	2460.2	952.23
39.555	53.574	2450.6	951.69
39.589	53.796	2440.3	951.06
39.623	54.021	2429.4	950.34
39.658	54.249	2417.9	949.55
39.692	54.48	2405.8	948.67
39.726	54.715	2393.2	947.71
39.76	54.952	2380.1	946.68
39.795	55.193	2366.6	945.57
39.829	55.438	2352.7	944.38
39.863	55.685	2338.5	943.12
39.898	55.937	2323.9	941.78
39.932	56.192	2308.9	940.38
39.967	56.45	2293.7	938.9
40.001	56.713	2278.2	937.35
40.036	56.979	2262.5	935.74
40.056	57.141	2252.9	934.74
40.066	57.219	2247.2	934.49
40.076	57.298	2241.6	934.24
40.09	57.406	2234.	933.88
40.124	57.681	2215.5	932.97
40.159	57.967	2197.	932.
40.195	58.265	2178.5	930.97
40.232	58.576	2159.9	929.88
40.27	58.899	2141.2	928.73
40.309	59.236	2122.6	927.5
40.349	59.586	2103.8	926.21
40.389	59.95	2085.1	924.84
40.43	60.329	2066.3	923.4
40.472	60.721	2047.6	921.87
40.515	61.128	2028.7	920.27
40.558	61.55	2009.9	918.59
40.602	61.988	1991.1	916.83
40.647	62.44	1972.2	914.98
40.692	62.908	1953.3	913.06

40.738	63.391	1934.4	911.04
40.784	63.891	1915.5	908.95
40.831	64.406	1896.5	906.77
40.878	64.938	1877.5	904.51
40.926	65.486	1858.6	902.17
40.974	66.05	1846.7	899.75
41.022	66.632	1840.8	897.24
41.058	67.066	1836.3	895.36
41.068	67.189	1835.1	894.98
41.078	67.313	1833.8	894.59
41.1	67.592	1830.8	893.72
41.144	68.144	1824.9	892.01
41.189	68.73	1818.5	890.22
41.236	69.351	1811.7	888.33
41.284	70.009	1804.5	886.35
41.334	70.706	1796.7	884.28
41.386	71.443	1788.5	882.1
41.44	72.222	1779.8	879.81
41.495	73.043	1770.6	877.42
41.552	73.908	1761.	874.91
41.61	74.818	1750.9	872.3
41.67	75.774	1740.5	869.57
41.731	76.776	1729.6	866.73
41.793	77.824	1718.4	863.78
41.856	78.918	1706.9	860.73
41.921	80.059	1695.	857.56
41.986	81.247	1682.9	854.29
42.023	81.925	1676.	852.43
42.059	82.613	1669.1	850.56
42.069	82.802	1667.3	850.14
42.079	82.991	1665.4	849.72
42.109	83.565	1659.6	848.46
42.17	84.752	1648.1	845.9
42.234	86.029	1635.8	843.2
42.301	87.399	1622.9	840.35
42.371	88.867	1609.4	837.35
42.444	90.435	1595.2	834.19
42.519	92.105	1580.5	830.88
42.598	93.878	1565.3	827.42
42.679	95.754	1549.7	823.8
42.762	97.732	1533.6	820.04
42.847	99.81	1517.2	816.14
42.934	101.98	1500.5	812.1
43.023	104.25	1483.6	807.93
43.061	105.23	1476.3	806.13
43.071	105.49	1474.6	805.72
43.081	105.75	1472.7	805.31
43.111	106.54	1466.9	804.08
43.143	107.39	1460.7	802.76
43.228	109.69	1444.7	799.28
43.318	112.17	1427.8	795.58
43.413	114.85	1410.2	791.67
43.513	117.73	1391.8	787.55
43.613	120.68	1373.6	783.39
43.714	123.69	1355.5	779.21
43.814	126.76	1337.7	774.99
43.914	129.89	1320.1	770.75
44.014	133.08	1302.7	766.47
44.062	134.62	1294.5	764.41
44.072	134.94	1292.9	764.03
44.082	135.26	1291.2	763.64

44.112	136.24	1286.	762.49
44.172	138.18	1275.8	760.22
44.272	141.5	1259.	756.41
44.372	144.87	1242.5	752.6
44.472	148.29	1226.2	748.79
44.572	151.75	1210.2	744.99
44.672	155.25	1194.3	741.17
44.772	158.8	1178.8	737.35
44.872	162.38	1163.5	733.51
44.973	165.99	1148.5	729.66
45.063	169.29	1135.1	726.16
45.073	169.66	1133.6	725.81
45.083	170.02	1132.2	725.45
45.113	171.12	1127.8	724.4
45.193	174.05	1123.5	721.61
45.293	177.76	1118.9	718.12
45.393	181.49	1114.5	714.65
45.493	185.25	1110.3	711.18
45.594	189.02	1106.5	707.72
45.694	192.81	1103.	704.26
45.794	196.61	1099.5	700.8
45.894	200.42	1096.1	697.34
45.994	204.23	1092.4	693.88
46.065	206.92	1089.8	691.44
46.075	207.31	1089.4	691.12
46.085	207.69	1089.	690.8
46.115	208.83	1087.9	689.83
46.205	212.27	1084.3	686.97
46.305	216.09	1080.3	683.8
46.405	219.92	1076.	680.65
46.505	223.74	1071.7	677.51
46.606	227.57	1067.3	674.38
46.706	231.39	1062.7	671.26
46.806	235.21	1058.	668.14
46.906	239.03	1053.3	665.03
47.006	242.83	1048.4	661.92
47.066	245.12	1045.5	660.06
47.076	245.5	1045.	659.77
47.086	245.88	1044.5	659.48
47.116	247.02	1043.	658.61
47.206	250.43	1038.5	656.01
47.307	254.21	1033.4	653.14
47.407	257.98	1028.3	650.29
47.507	261.74	1023.1	647.45
47.607	265.48	1017.8	644.63
47.707	269.21	1012.5	641.81
47.807	272.92	1007.1	639.01
47.907	276.61	1001.7	636.21
48.008	280.29	996.22	633.41
48.068	282.49	992.93	631.74
48.078	282.85	992.38	631.48
48.088	283.22	991.83	631.21
48.118	284.31	990.18	630.42
48.208	287.58	985.2	628.07
48.308	291.2	979.64	625.48
48.408	294.79	974.05	622.9
48.508	298.35	968.45	620.34
48.608	301.89	962.82	617.79
48.708	305.41	957.17	615.25
48.809	308.9	951.52	612.71
48.909	312.36	945.85	610.19

49.009	315.8	940.17	607.68
49.069	317.85	936.77	606.17
49.079	318.19	936.2	605.93
49.089	318.53	935.63	605.69
49.119	319.55	933.93	604.98
49.209	322.59	928.81	602.85
49.309	325.94	923.13	600.51
49.409	329.26	917.45	598.18
49.51	332.55	911.79	595.86
49.61	335.81	906.12	593.56
49.71	339.04	900.47	591.26
49.81	342.23	894.83	588.98
49.91	345.39	889.2	586.71
50.01	348.52	883.58	584.44
50.07	350.39	880.22	583.08
50.08	350.7	879.66	582.87
50.09	351.01	879.1	582.65
50.12	351.94	877.42	582.01
50.211	354.69	872.39	580.08
50.311	357.72	866.82	577.96
50.411	360.72	861.28	575.85
50.511	363.68	855.75	573.76
50.611	366.6	850.25	571.68
50.711	369.49	844.77	569.61
50.811	372.35	839.32	567.55
50.912	375.17	833.89	565.5
51.012	377.96	828.49	563.46
51.072	379.62	825.26	562.24
51.082	379.89	824.72	562.04
51.092	380.17	824.18	561.85
51.122	380.99	822.57	561.26
51.212	383.44	817.76	559.52
51.312	386.12	812.45	557.61
51.412	388.76	807.17	555.71
51.512	391.37	801.92	553.82
51.613	393.94	796.7	551.94
51.713	396.48	791.51	550.07
51.813	398.98	786.36	548.22
51.913	401.45	781.24	546.37
52.013	403.88	776.16	544.53
52.073	405.32	773.12	543.43
52.083	405.56	772.61	543.26
52.093	405.8	772.11	543.08
52.123	406.52	770.6	542.55
52.213	408.65	766.09	540.99
52.314	410.97	761.11	539.26
52.414	413.26	756.17	537.55
52.514	415.51	751.27	535.85
52.614	417.73	746.41	534.16
52.714	419.92	741.58	532.48
52.814	418.9	736.79	530.81
52.914	416.67	732.05	529.15
53.015	414.44	727.34	527.5
53.075	413.1	724.52	526.51
53.085	412.	724.06	526.23
53.095	410.9	723.59	525.96
53.125	407.63	722.19	525.13
53.215	398.04	718.02	522.52
53.315	387.51	713.43	519.52
53.415	377.04	708.88	516.43
53.515	366.61	704.36	513.25

53.615	356.21	699.89	509.99
53.716	345.83	695.45	506.66
53.816	335.47	691.05	503.27
53.916	325.12	686.68	499.81
54.016	314.77	682.34	496.28
54.076	308.57	679.74	494.15
54.086	307.54	679.31	493.8
54.096	306.5	678.88	493.44
54.126	303.4	677.58	492.36
54.216	294.09	673.72	489.09
54.316	283.75	669.45	485.4
54.417	273.41	665.19	481.66
54.517	263.08	660.93	477.87
54.617	252.76	656.69	474.04
54.717	242.43	652.44	470.15
54.817	232.11	648.18	466.23
54.917	221.8	643.92	462.26
55.017	211.48	639.64	458.24
55.078	205.3	637.07	455.82
55.088	205.2	636.64	455.53
55.098	205.08	636.21	455.24
55.128	204.67	634.92	454.37
55.215	203.25	631.18	451.93
55.315	201.48	626.84	449.2
55.415	199.64	622.48	446.54
55.515	197.77	618.09	443.94
55.615	195.86	613.66	441.39
55.715	193.93	609.21	438.87
55.815	191.99	604.73	436.4
55.916	190.03	600.21	433.97
56.016	188.07	595.66	431.57
56.079	186.82	592.78	430.06
56.089	186.85	592.33	429.85
56.099	186.87	591.87	429.64
56.129	186.94	590.5	429.02
56.219	187.07	586.36	427.2
56.319	187.18	581.74	425.23
56.419	187.27	577.1	423.31
56.52	187.35	572.62	421.43
56.62	187.41	569.44	419.6
56.72	187.47	566.38	417.81
56.82	187.52	563.42	416.06
56.92	187.56	560.57	414.34
57.02	187.6	557.81	412.66
57.08	187.62	556.18	411.66
57.09	187.63	555.91	411.5
57.1	187.64	555.65	411.33
57.13	187.66	554.88	410.84
57.22	187.71	552.55	409.38
57.321	187.77	550.09	407.8
57.421	187.83	547.7	406.24
57.521	187.88	545.38	404.71
57.621	187.93	543.2	403.21
57.721	187.98	541.11	401.74
57.821	188.02	539.08	400.28
57.921	188.07	537.16	398.86
58.022	188.11	535.34	397.45
58.082	188.13	534.26	396.62
58.092	188.14	534.08	396.48
58.102		533.92	396.34
58.132	188.15	533.42	395.93

58.222	188.19	531.93	394.7
58.322	188.22	530.39	393.36
58.422	188.26	528.92	392.04
58.522	188.29	527.54	390.74
58.622	188.32	526.26	389.46
58.723	188.36	525.07	388.2
58.823	188.39	523.97	386.95
58.923	188.41	522.95	385.73
59.023	188.44	522.01	384.52
59.083	188.46	521.5	383.8
59.093	188.41	521.42	383.67
59.103	188.35	521.34	383.54
59.133	188.2	521.09	383.16
59.223	187.74	520.4	382.03
59.323	187.24	519.71	380.78
59.424	186.75	519.07	379.54
59.524	186.25	518.51	378.31
59.624	185.76	517.99	377.09
59.724	185.27	517.51	375.87
59.824	184.78	517.06	374.67
59.924	184.29	516.63	373.47
60.024	183.79	516.21	372.29
60.084	183.5	515.97	371.58
60.095	183.45	515.93	371.46
60.105	183.4	515.88	371.34
60.135	183.24	515.76	370.99
60.225	182.77	515.38	369.93
60.325	182.24	514.95	368.76
60.425	181.72	514.52	367.6
60.525	181.2	514.08	366.45
60.625	180.67	513.62	365.31
60.725	180.15	513.17	364.17
60.826	179.62	512.71	363.05
60.926	179.1	512.23	361.92
61.026	178.57	511.74	360.81
61.086	178.26	511.44	360.14
61.096	178.21	511.39	360.03
61.106	178.17	511.34	359.92
61.136	178.04	511.19	359.6
61.226	177.63	510.74	358.62
61.326	177.18	510.22	357.54
61.426	176.73	509.69	356.46
61.527	176.28	509.15	355.4
61.627	175.83	508.6	354.34
61.727	175.38	508.04	353.29
61.827	174.93	507.47	352.24
61.927	174.48	506.89	351.2
62.027	174.03	506.3	350.17
62.087	173.76	505.94	349.55
62.097	173.72	505.88	349.45
62.107	173.68	505.82	349.35
62.137	173.57	505.64	349.05
62.227	173.24	505.09	348.14
62.328	172.87	504.47	347.14
62.428	172.5	503.84	346.15
62.528	172.13	503.21	345.16
62.628	171.75	502.57	344.18
62.728	171.38	501.91	343.21
62.828	171.01	501.25	342.24
62.928	170.63	500.58	341.28
63.029	170.26	499.91	340.32

63.089	170.03	499.5	339.75
63.099	170.	499.43	339.66
63.109	169.97	499.36	339.56
63.139	169.89	499.16	339.28
63.229	169.62	498.54	338.44
63.329	169.32	497.84	337.52
63.429	169.03	497.14	336.6
63.529	168.73	496.43	335.69
63.629	168.43	495.72	334.78
63.73	168.13	495.	333.88
63.83	167.83	494.28	332.98
63.93	167.53	493.55	332.09
64.03	167.23	492.82	331.21
64.09	167.05	492.38	330.68
64.1	167.03	492.31	330.6
64.11	167.01	492.24	330.51
64.14	166.94	492.02	330.25
64.23	166.74	491.35	329.47
64.33	166.51	490.71	328.62
64.431	166.28	490.19	327.77
64.531	166.05	489.67	326.92
64.631	165.82	489.14	326.08
64.731	165.59	488.6	325.25
64.831	165.36	488.05	324.42
64.931	165.12	487.49	323.6
65.031	164.89	486.92	322.78
65.092	164.75	486.58	322.29
65.102	164.74	486.52	322.21
65.112	164.72	486.46	322.13
65.142	164.67	486.29	321.89
65.232	164.52	485.76	321.17
65.332	164.35	485.17	320.38
65.432	164.18	484.57	319.59
65.532	164.	483.95	318.81
65.632	163.83	483.34	318.03
65.732	163.66	482.71	317.26
65.833	163.49	482.08	316.49
65.933	163.31	481.43	315.73
66.033	163.14	480.79	314.97
66.093	163.04	480.4	314.52
66.103	163.03	480.33	314.45
66.113	163.01	480.26	314.37
66.143	162.98	480.07	314.15
66.233	162.87	479.47	313.48
66.333	162.74	478.8	312.75
66.433	162.62	478.13	312.02
66.534	162.5	477.45	311.29
66.634	162.37	476.76	310.57
66.734	162.25	476.07	309.86
66.834	162.12	475.37	309.14
66.934	162.	474.66	308.44
67.034	161.87	473.95	307.73
67.094	161.8	473.53	307.32
67.104	161.79	473.46	307.25
67.114	161.78	473.38	307.18
67.144	161.76	473.17	306.97
67.235	161.68	472.52	306.35
67.335	161.6	471.8	305.67
67.435	161.51	471.07	304.99
67.535	161.42	470.33	304.31
67.635	161.34	469.59	303.65

67.735	161.25	468.85	302.98
67.835	161.16	468.1	302.32
67.936	161.08	467.35	301.66
68.036	160.99	466.59	301.01
68.096	160.94	466.13	300.62
68.106	160.93	466.06	300.55
68.116		465.98	300.49
68.146	160.91	465.75	300.3
68.236	160.86	465.07	299.72
68.336	160.8	464.3	299.08
68.436	160.74	463.53	298.45
68.536	160.68	462.75	297.82
68.637	160.63	461.97	297.2
68.737	160.57	461.19	296.58
68.837	160.51	460.41	295.96
68.937	160.45	459.62	295.35
69.037	160.39	458.83	294.74
69.097	160.36	458.36	294.37
69.107	160.35	458.28	294.31
69.117		458.2	294.25
69.147	160.34	457.96	294.07
69.237	160.31	457.24	293.53
69.338	160.27	456.44	292.94
69.438	160.23	455.64	292.35
69.538	160.19	454.84	291.76
69.638	160.16	454.04	291.18
69.738	160.12	453.23	290.6
69.838	160.08	452.42	290.02
69.938	160.04	451.61	289.45
70.039	160.01	450.79	288.88
70.099	159.98	450.3	288.54
70.109		450.22	288.48
70.119		450.14	288.42
70.149	159.97	449.9	288.25
70.239	159.95	449.16	287.75
70.339	159.93	448.34	287.19
70.439	159.91	447.52	286.64
70.539	159.88	446.7	286.09
70.639	159.86	445.88	285.54
70.739	159.84	445.05	284.99
70.84	159.81	444.23	284.45
70.94	159.79	443.4	283.91
71.04	159.77	442.57	283.38
71.1	159.75	442.08	283.06

FIGURE 6
Model (A4) > Transient Thermal (A5) > Solution (A6) > Total Heat Flux

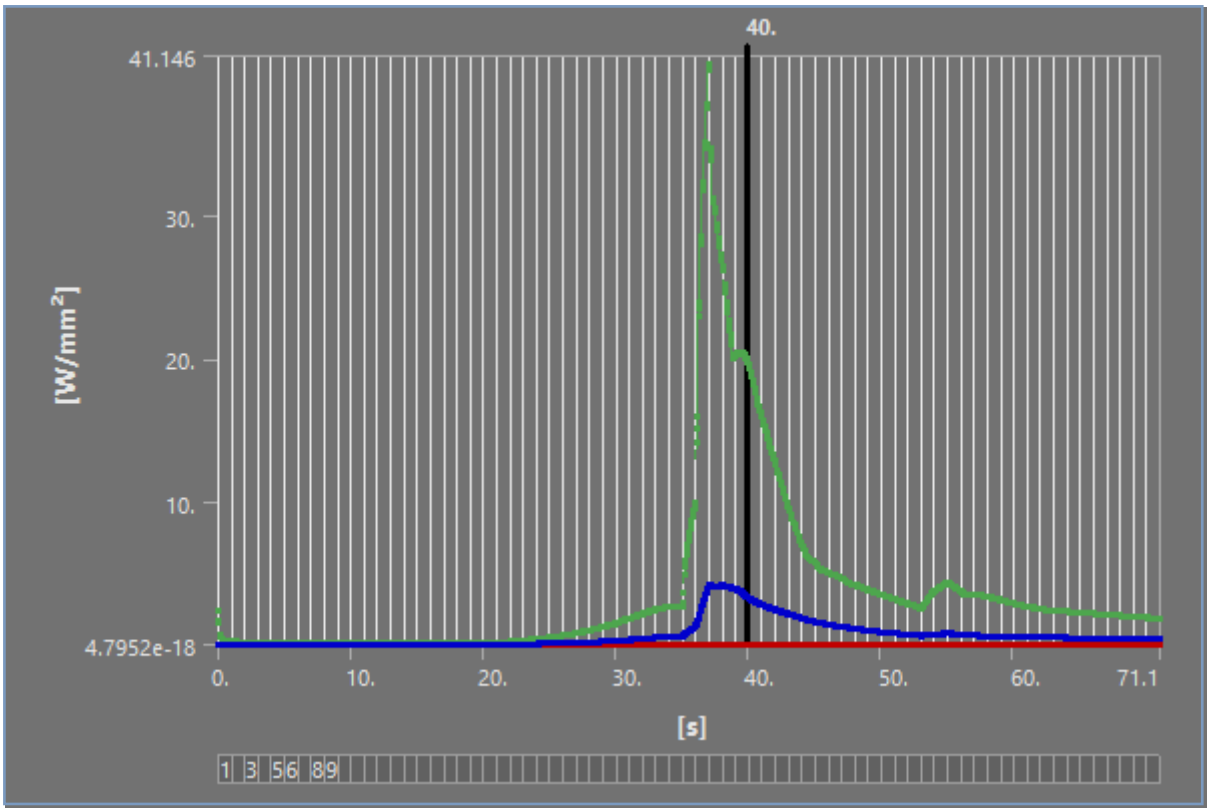


TABLE 24
Model (A4) > Transient Thermal (A5) > Solution (A6) > Total Heat Flux

Time [s]	Minimum [W/mm²]	Maximum [W/mm²]	Average [W/mm²]
1.0014e-002	1.1277e-016	2.4103	5.7666e-002
1.9254e-002	1.2108e-016	1.7383	4.8201e-002
2.5346e-002	1.3273e-015	1.4881	4.3953e-002
3.1439e-002	3.3749e-015	1.3183	4.1085e-002
4.7761e-002	2.626e-015	1.0727	3.7544e-002
7.4673e-002	7.2779e-016	0.86043	3.4974e-002
0.11933	1.5035e-016	0.68059	3.2833e-002
0.19327	3.0489e-016	0.53001	3.1145e-002
0.27247	3.0771e-016	0.43311	3.0128e-002
0.37261	5.6831e-015	0.36305	2.9258e-002
0.47275	6.825e-014	0.31998	2.8583e-002
0.57289	3.1537e-013	0.28921	2.8053e-002
0.67303	1.1337e-012	0.26549	2.7633e-002
0.77317	3.3582e-012	0.251	2.7295e-002
0.87331	7.6404e-012	0.24644	2.7024e-002
0.93735	1.1698e-011	0.24296	2.6874e-002
1.0014	1.7593e-011	0.23912	2.6743e-002
1.0114	1.8568e-011	0.23849	2.6723e-002
1.0214	1.9594e-011	0.23784	2.6704e-002
1.0515	2.3195e-011	0.23585	2.665e-002
1.1416	4.1707e-011	0.22974	2.6516e-002
1.2417	8.1152e-011	0.22293	2.6395e-002
1.3419	1.5819e-010	0.21622	2.6295e-002
1.442	2.9598e-010	0.20968	2.6212e-002
1.5422	4.3173e-010	0.20333	2.6143e-002
1.6423	5.7922e-010	0.19721	2.6084e-002
1.7424	7.1892e-010	0.19131	2.6033e-002
1.8426	7.761e-010	0.18566	2.5988e-002
1.9427	1.0302e-009	0.18024	2.5947e-002
2.0028	1.343e-009	0.17707	2.5923e-002
2.0128	1.4076e-009	0.17653	2.5919e-002
2.0228	1.4704e-009	0.17601	2.5915e-002
2.0529	1.6313e-009	0.17445	2.5904e-002

2.143	1.7878e-009	0.16994	2.5871e-002
2.2431	1.5668e-010	0.16516	2.5835e-002
2.3433	1.4061e-009	0.16059	2.58e-002
2.4434	1.1235e-009	0.15622	2.5765e-002
2.5436	6.1234e-010	0.15205	2.573e-002
2.6437	8.8719e-010	0.14805	2.5694e-002
2.7438	2.4442e-010	0.14423	2.5657e-002
2.844	3.738e-010	0.14058	2.5619e-002
2.9441	1.3905e-010	0.13707	2.5581e-002
3.0042	1.5062e-010	0.13503	2.5557e-002
3.0142	1.3934e-010	0.13468	2.5553e-002
3.0242	1.2885e-010	0.13435	2.5549e-002
3.0543	1.0219e-010	0.13335	2.5537e-002
3.1444	6.1556e-011	0.13353	2.55e-002
3.2445	6.6752e-011	0.13408	2.5458e-002
3.3447	8.5803e-011	0.13473	2.5415e-002
3.4448	9.9749e-011	0.13527	2.5372e-002
3.545	8.9079e-011	0.1357	2.5327e-002
3.6451	4.6708e-011	0.13603	2.5282e-002
3.7452	1.7901e-011	0.13627	2.5236e-002
3.8454	8.6572e-012	0.13643	2.519e-002
3.9455	1.8992e-011	0.13651	2.5143e-002
4.0056	9.3227e-012	0.13653	2.5114e-002
4.0156	1.0241e-011		2.5109e-002
4.0256	1.2055e-011		2.5104e-002
4.0557	1.8735e-011		2.509e-002
4.1458	1.2994e-011	0.13647	2.5047e-002
4.2459	7.673e-012	0.13634	2.4998e-002
4.3461	4.3558e-012	0.13616	2.4949e-002
4.4462	3.1205e-012	0.13592	2.4899e-002
4.5464	3.4473e-012	0.13564	2.4849e-002
4.6465	4.0781e-012	0.13531	2.4799e-002
4.7466	3.4004e-012	0.13495	2.4749e-002
4.8468	2.503e-012	0.13454	2.4698e-002
4.9469	2.1358e-012	0.1341	2.4648e-002
5.007	2.0692e-012	0.13383	2.4617e-002
5.017	2.0665e-012	0.13378	2.4612e-002
5.027	2.0653e-012	0.13373	2.4607e-002
5.0571	2.0651e-012	0.13359	2.4592e-002
5.1472	2.0564e-012	0.13313	2.4546e-002
5.2474	2.0233e-012	0.1326	2.4494e-002
5.3475	1.9526e-012	0.13205	2.4443e-002
5.4477	1.8483e-012	0.13147	2.4391e-002
5.5478	1.6175e-012	0.13087	2.434e-002
5.648	1.3576e-012	0.13026	2.4288e-002
5.7481	1.1515e-012	0.12963	2.4237e-002
5.8483	9.5013e-013	0.12898	2.4185e-002
5.9484	6.085e-013	0.12832	2.4133e-002
6.0085	4.583e-013	0.12792	2.4102e-002
6.0185	4.3634e-013	0.12785	2.4097e-002
6.0285	4.1592e-013	0.12779	2.4091e-002
6.0586	3.6519e-013	0.12758	2.4076e-002
6.1487	2.8391e-013	0.12697	2.4029e-002
6.2488	2.6267e-013	0.12627	2.3977e-002
6.349	1.2119e-013	0.12557	2.3925e-002
6.4491	9.1129e-015	0.12486	2.3873e-002
6.5493	1.0421e-013	0.12415	2.3821e-002
6.6494	6.9825e-014	0.12343	2.3768e-002
6.7495	5.0303e-014	0.12271	2.3716e-002
6.8497	1.4295e-014	0.12198	2.3664e-002
6.9498	4.1034e-014	0.12125	2.3611e-002

7.0099	4.1509e-014	0.12081	2.358e-002
7.0199	4.1674e-014	0.12074	2.3575e-002
7.0299	4.1837e-014	0.12066	2.357e-002
7.06	4.2191e-014	0.12044	2.3554e-002
7.1501	4.2358e-014	0.11978	2.3507e-002
7.2502	4.1555e-014	0.11905	2.3454e-002
7.3504	2.473e-014	0.11831	2.3402e-002
7.4505	2.2185e-014	0.11758	2.3349e-002
7.5507	2.2569e-014	0.11684	2.3296e-002
7.6508	1.7095e-014	0.11611	2.3244e-002
7.7509	1.2865e-014	0.11573	2.3191e-002
7.8511	9.5977e-015	0.11537	2.3138e-002
7.9512	7.0577e-015	0.11501	2.3086e-002
8.0113	3.8622e-015	0.1148	2.3054e-002
8.0213	3.3956e-015	0.11476	2.3049e-002
8.0313	2.9839e-015	0.11472	2.3043e-002
8.0614	2.376e-015	0.11461	2.3028e-002
8.1515	3.4352e-015	0.11428	2.298e-002
8.2516	2.3103e-015	0.11391	2.2927e-002
8.3518	1.4448e-015	0.11354	2.2874e-002
8.4519	7.8302e-016	0.11316	2.2821e-002
8.5521	3.947e-016	0.11279	2.2768e-002
8.6522	4.8037e-016	0.11241	2.2715e-002
8.7523	7.4415e-016	0.11203	2.2662e-002
8.8525	9.7571e-016	0.11165	2.2608e-002
8.9526	1.1336e-015	0.11127	2.2555e-002
9.0127	1.2437e-015	0.11104	2.2523e-002
9.0227	1.2715e-015	0.111	2.2518e-002
9.0327	1.29e-015	0.11096	2.2513e-002
9.0628	1.3843e-015	0.11085	2.2497e-002
9.1529	1.4798e-015	0.1105	2.2449e-002
9.253	1.1799e-015	0.11011	2.2395e-002
9.3532	6.6853e-016	0.10973	2.2342e-002
9.4533	3.0732e-016	0.10934	2.2288e-002
9.5535	3.8569e-016	0.10896	2.2235e-002
9.6536	6.4427e-016	0.10858	2.2181e-002
9.7537	6.5026e-016	0.10819	2.2128e-002
9.8539	5.5898e-016	0.10781	2.2074e-002
9.954	2.6664e-016	0.10743	2.202e-002
10.014	1.2085e-016	0.1072	2.1988e-002
10.024	6.7922e-017	0.10716	2.1983e-002
10.034	3.599e-017	0.10712	2.1977e-002
10.064	3.3314e-017	0.10701	2.1961e-002
10.154	1.7662e-016	0.10666	2.1913e-002
10.254	1.2205e-016	0.10628	2.1859e-002
10.355	8.1318e-017	0.1059	2.1805e-002
10.455	8.5063e-017	0.10553	2.1751e-002
10.555	6.2649e-017	0.10515	2.1697e-002
10.655	2.9322e-017	0.10478	2.1643e-002
10.755	6.0499e-017	0.1044	2.159e-002
10.855	4.2255e-017	0.10403	2.1536e-002
10.955	4.4074e-017	0.10366	2.1482e-002
11.016	6.193e-017	0.10344	2.1449e-002
11.026	5.7414e-017	0.1034	2.1444e-002
11.036	3.8358e-017	0.10336	2.1438e-002
11.066	5.3069e-017	0.10325	2.1422e-002
11.156	2.6009e-017	0.10292	2.1374e-002
11.256	3.7914e-017	0.10255	2.132e-002
11.356	3.071e-017	0.10218	2.1265e-002
11.456	2.3245e-017	0.10182	2.1211e-002
11.556	3.9961e-017	0.10146	2.1157e-002

11.656	3.2912e-017	0.1011	2.1103e-002
11.757	3.1989e-017	0.10074	2.1049e-002
11.857	1.7104e-017	0.10038	2.0995e-002
11.957	2.8337e-017	0.10003	2.0941e-002
12.017	2.4701e-017	9.9813e-002	2.0908e-002
12.027	3.8072e-017	9.9776e-002	2.0903e-002
12.037	7.8691e-018	9.9741e-002	2.0898e-002
12.067	2.5388e-017	9.9634e-002	2.0881e-002
12.157	2.8279e-017	9.9316e-002	2.0833e-002
12.257	2.4176e-017	9.8964e-002	2.0779e-002
12.357	2.6662e-017	9.8615e-002	2.0725e-002
12.458	2.6606e-017	9.8268e-002	2.067e-002
12.558	2.4068e-017	9.7922e-002	2.0616e-002
12.658	2.0847e-017	9.7578e-002	2.0562e-002
12.758	2.4913e-017	9.7236e-002	2.0508e-002
12.858	2.8595e-017	9.6896e-002	2.0454e-002
12.958	2.405e-017	9.6558e-002	2.04e-002
13.018	1.8384e-017	9.6355e-002	2.0368e-002
13.028	4.4639e-017	9.6321e-002	2.0362e-002
13.038	3.5098e-017	9.6287e-002	2.0357e-002
13.068	3.3568e-017	9.6185e-002	2.0341e-002
13.158	3.8678e-017	9.5882e-002	2.0292e-002
13.259	2.4542e-017	9.5548e-002	2.0238e-002
13.359	2.856e-017	9.5216e-002	2.0184e-002
13.459	3.2556e-017	9.4885e-002	2.013e-002
13.559	3.3273e-017	9.4557e-002	2.0076e-002
13.659	3.6946e-017	9.423e-002	2.0022e-002
13.759	3.5647e-017	9.3904e-002	1.9969e-002
13.859	3.4481e-017	9.3581e-002	1.9915e-002
13.96	3.3918e-017	9.3259e-002	1.9861e-002
14.02	4.3492e-017	9.3067e-002	1.9829e-002
14.03	4.5211e-017	9.3034e-002	1.9823e-002
14.04	4.6926e-017	9.3002e-002	1.9818e-002
14.07	3.5934e-017	9.2905e-002	1.9802e-002
14.16	3.7248e-017	9.2617e-002	1.9754e-002
14.26	3.3334e-017	9.2299e-002	1.97e-002
14.36	3.29e-017	9.1983e-002	1.9646e-002
14.46	2.2415e-017	9.1669e-002	1.9593e-002
14.56	2.0383e-017	9.1357e-002	1.9539e-002
14.661	9.9981e-018	9.1046e-002	1.9486e-002
14.761	2.7226e-017	9.0737e-002	1.9432e-002
14.861	2.1019e-017	9.0429e-002	1.9379e-002
14.961	2.181e-017	9.0123e-002	1.9326e-002
15.021	1.0691e-017	8.9939e-002	1.9294e-002
15.031	3.444e-017	8.9908e-002	1.9288e-002
15.041	2.6408e-017	8.9878e-002	1.9283e-002
15.071	2.9198e-017	8.9786e-002	1.9267e-002
15.161	2.9914e-017	8.9512e-002	1.9219e-002
15.261	2.5108e-017	8.9209e-002	1.9166e-002
15.362	1.6745e-017	8.8909e-002	1.9113e-002
15.462	1.0956e-017	8.861e-002	1.906e-002
15.562	1.5163e-017	8.8312e-002	1.9007e-002
15.662	1.3231e-017	8.8016e-002	1.8954e-002
15.762	1.5389e-017	8.7721e-002	1.8901e-002
15.862	1.6297e-017	8.7428e-002	1.8848e-002
15.962	1.5391e-017	8.7136e-002	1.8795e-002
16.022	2.2129e-017	8.6962e-002	1.8764e-002
16.033	2.3862e-017	8.6932e-002	1.8758e-002
16.043	2.104e-017	8.6903e-002	1.8753e-002
16.073	2.1641e-017	8.6815e-002	1.8737e-002
16.163	2.1501e-017	8.6554e-002	1.869e-002

16.263	1.9215e-017	8.6266e-002	1.8638e-002
16.363	1.6159e-017	8.5979e-002	1.8585e-002
16.463	1.5065e-017	8.5694e-002	1.8533e-002
16.563	4.7952e-018	8.541e-002	1.848e-002
16.663	1.8499e-017	8.5127e-002	1.8428e-002
16.764	1.6058e-017	8.4846e-002	1.8376e-002
16.864	1.5018e-017	8.4566e-002	1.8324e-002
16.964	1.8836e-017	8.4287e-002	1.8272e-002
17.024	1.3928e-017	8.412e-002	1.8241e-002
17.034	3.764e-012	8.4092e-002	1.8237e-002
17.044	6.492e-012	8.4064e-002	1.8232e-002
17.074	9.9547e-012	8.3981e-002	1.822e-002
17.164	6.3049e-012	8.3731e-002	1.8181e-002
17.264	1.1859e-011	8.3456e-002	1.8138e-002
17.364	1.2496e-011	8.3182e-002	1.8094e-002
17.465	1.341e-011	8.2911e-002	1.8051e-002
17.565	1.7113e-011	8.2641e-002	1.8007e-002
17.665	2.1501e-011	8.2374e-002	1.7963e-002
17.765	2.4883e-011	8.2109e-002	1.7919e-002
17.865	2.71e-011	8.1846e-002	1.7876e-002
17.965	2.5244e-011	8.1587e-002	1.7832e-002
18.025	2.1471e-011	8.1432e-002	1.7806e-002
18.035	6.4251e-011	8.1406e-002	1.7827e-002
18.045	3.0803e-011	8.138e-002	1.7846e-002
18.075	3.9849e-011	8.1303e-002	1.7898e-002
18.165	1.8881e-010	8.1075e-002	1.8039e-002
18.265	2.2972e-010	8.0828e-002	1.8187e-002
18.365	2.3961e-010	8.0591e-002	1.8329e-002
18.466	2.6156e-010	8.037e-002	1.8466e-002
18.566	3.5575e-010	8.017e-002	1.8599e-002
18.666	4.6551e-010	7.9996e-002	1.873e-002
18.766	5.527e-010	7.9853e-002	1.8858e-002
18.866	6.1552e-010	7.9745e-002	1.8985e-002
18.966	5.8389e-010	7.9674e-002	1.911e-002
19.027	4.8705e-010	7.9645e-002	1.9185e-002
19.037	4.7496e-010	7.964e-002	1.9234e-002
19.047	7.3243e-010	7.9636e-002	1.928e-002
19.077	8.514e-010	7.9628e-002	1.9409e-002
19.167	6.2867e-010	7.9634e-002	1.9777e-002
19.267	5.3916e-010	7.9685e-002	2.0173e-002
19.367	6.3466e-010	7.9784e-002	2.0559e-002
19.467	7.9904e-010	7.9936e-002	2.0939e-002
19.568	8.8343e-010	8.015e-002	2.1312e-002
19.668	1.2825e-009	8.0431e-002	2.1682e-002
19.768	1.4534e-009	8.0783e-002	2.2048e-002
19.868	1.0909e-009	8.121e-002	2.2412e-002
19.968	3.5792e-010	8.3853e-002	2.2774e-002
20.028	3.8012e-010	8.5442e-002	2.299e-002
20.038	9.8491e-010	8.7202e-002	2.3064e-002
20.048	1.2529e-009	8.8709e-002	2.3135e-002
20.078	4.4769e-010	9.2195e-002	2.3339e-002
20.168	1.3661e-009	9.966e-002	2.3932e-002
20.269	2.4615e-009	0.10638	2.4576e-002
20.369	2.7621e-009	0.11222	2.521e-002
20.469	7.2862e-010	0.11749	2.5837e-002
20.569	1.831e-009	0.12236	2.6457e-002
20.669	1.6266e-009	0.1269	2.7074e-002
20.769	5.3143e-010	0.13119	2.7686e-002
20.869	1.156e-009	0.13527	2.8296e-002
20.97	8.8965e-010	0.13916	2.8903e-002
21.03	8.4717e-010	0.14144	2.9267e-002

21.04	1.1025e-009	0.14341	2.9368e-002
21.05	8.6389e-010	0.14512	2.9466e-002
21.08	1.5733e-009	0.14912	2.9752e-002
21.17	1.2243e-009	0.15794	3.0586e-002
21.27	2.4781e-009	0.16603	3.1497e-002
21.37	2.2436e-009	0.17317	3.2398e-002
21.47	6.6157e-010	0.1797	3.3291e-002
21.57	1.1591e-009	0.18577	3.4178e-002
21.67	1.268e-009	0.19149	3.506e-002
21.771	1.6098e-009	0.19692	3.5938e-002
21.871	1.2381e-009	0.20211	3.6813e-002
21.971	2.28e-010	0.20708	3.7685e-002
22.031	5.8517e-010	0.21	3.8208e-002
22.041	1.1904e-009	0.21223	3.834e-002
22.051	1.7877e-009	0.21416	3.8469e-002
22.081	1.3703e-009	0.21875	3.8845e-002
22.171	2.7226e-009	0.22899	3.9947e-002
22.271	2.8578e-009	0.23853	4.1157e-002
22.371	2.486e-009	0.24704	4.2354e-002
22.472	7.7844e-010	0.25486	4.3544e-002
22.572	2.0381e-009	0.26218	4.4726e-002
22.672	2.1721e-009	0.26911	4.5903e-002
22.772	1.764e-009	0.27572	4.7076e-002
22.872	3.0728e-010	0.28205	4.8246e-002
22.972	1.0939e-009	0.28816	4.9412e-002
23.032	8.8672e-010	0.29174	5.0112e-002
23.042	1.2841e-009	0.29428	5.0279e-002
23.052	1.1703e-009	0.29649	5.0443e-002
23.082	1.8041e-009	0.30176	5.0922e-002
23.173	2.2155e-009	0.31367	5.2332e-002
23.273	2.3717e-009	0.32485	5.3882e-002
23.373	1.8115e-009	0.33489	5.5418e-002
23.473	9.6429e-010	0.34418	5.6946e-002
23.573	2.2901e-009	0.3529	5.8466e-002
23.673	2.1342e-009	0.36119	5.998e-002
23.773	7.6001e-010	0.36912	6.149e-002
23.874	1.3219e-009	0.37675	6.2996e-002
23.974	8.2861e-010	0.38412	6.4499e-002
24.034	5.3119e-010	0.38845	6.5401e-002
24.044	1.0221e-009	0.39138	6.561e-002
24.054	1.0868e-009	0.39394	6.5814e-002
24.084	1.7882e-009	0.40006	6.6415e-002
24.174	2.5726e-009	0.41398	6.8186e-002
24.274	2.4264e-009	0.42714	7.0134e-002
24.374	2.8918e-009	0.43901	7.2067e-002
24.474	1.2049e-009	0.45004	7.3991e-002
24.575	2.5546e-009	0.46044	7.5907e-002
24.675	1.5402e-009	0.47034	7.7816e-002
24.775	1.6888e-009	0.47985	7.9721e-002
24.875	1.1464e-009	0.48901	8.1622e-002
24.975	5.7711e-010	0.49788	8.352e-002
25.035	2.336e-010	0.50311	8.4659e-002
25.045	8.5291e-010	0.50652	8.4917e-002
25.055	1.3459e-009	0.50953	8.5171e-002
25.085	1.9733e-009	0.5167	8.5917e-002
25.175	2.8832e-009	0.53312	8.812e-002
25.276	3.1406e-009	0.54871	9.0545e-002
25.376	3.1698e-009	0.56285	9.2955e-002
25.476	1.5018e-009	0.57601	9.5353e-002
25.576	2.713e-009	0.58846	9.7743e-002
25.676	1.5791e-009	0.60036	0.10013

25.776	1.6921e-009	0.6118	0.10251
25.876	1.0613e-009	0.62286	0.10488
25.977	4.214e-010	0.63358	0.10725
26.037	1.8495e-010	0.63991	0.10868
26.047	7.9553e-010	0.64394	0.109
26.057	1.3758e-009	0.64751	0.10931
26.087	2.232e-009	0.65602	0.11023
26.177	3.2428e-009	0.67559	0.11297
26.277	3.5556e-009	0.69426	0.11597
26.377	3.5843e-009	0.71107	0.11896
26.477	1.8825e-009	0.72658	0.12193
26.577	3.0265e-009	0.74097	0.12488
26.677	1.4102e-009	0.75469	0.12783
26.778	1.8145e-009	0.76787	0.13077
26.878	1.0789e-009	0.78052	0.1337
26.978	4.051e-010	0.79261	0.13663
27.038	4.0392e-010	0.79927	0.13838
27.048	8.5123e-010	0.80435	0.13878
27.058	1.5182e-009	0.80875	0.13916
27.088	2.3777e-009	0.8192	0.14029
27.178	3.7713e-009	0.84263	0.14363
27.278	4.1501e-009	0.86457	0.14731
27.378	4.1795e-009	0.88427	0.15097
27.479	2.3409e-009	0.9025	0.15461
27.579	3.5004e-009	0.91968	0.15824
27.679	2.0039e-009	0.93604	0.16186
27.779	2.0518e-009	0.95172	0.16546
27.879	1.1874e-009	0.96575	0.16907
27.979	4.9136e-010	0.98008	0.17267
28.039	6.3081e-010	0.98855	0.17483
28.049	9.8497e-010	0.99469	0.1753
28.059	1.7507e-009	1.	0.17577
28.089	2.8768e-009	1.0127	0.17715
28.18	4.4547e-009	1.0413	0.18125
28.28	4.9095e-009	1.0681	0.18576
28.38	4.9389e-009	1.0923	0.19024
28.48	2.8472e-009	1.1147	0.19471
28.58	4.1126e-009	1.1358	0.19916
28.68	2.8606e-009	1.1558	0.20359
28.78	2.3753e-009	1.1735	0.208
28.881	1.3537e-009	1.1912	0.21241
28.981	6.2512e-010	1.2087	0.21682
29.041	8.7298e-010	1.219	0.21947
29.051	1.1547e-009	1.2263	0.22006
29.061	2.0219e-009	1.2327	0.22063
29.091	3.6514e-009	1.2478	0.22232
29.181	5.2049e-009	1.2821	0.22731
29.281	5.7327e-009	1.3145	0.23282
29.381	5.7513e-009	1.3437	0.2383
29.481	3.2598e-009	1.3708	0.24375
29.582	4.7407e-009	1.3964	0.2492
29.682	3.8014e-009	1.419	0.25462
29.782	2.6718e-009	1.4414	0.26004
29.882	1.4898e-009	1.4633	0.26546
29.982	8.3593e-010	1.4846	0.27086
30.042	1.2419e-009	1.4973	0.2741
30.052	1.3219e-009	1.5057	0.27481
30.062	2.2044e-009	1.5131	0.2755
30.092	4.0054e-009	1.5306	0.27754
30.182	5.7417e-009	1.5707	0.28358
30.283	6.3029e-009	1.6085	0.29023

30.383	6.2742e-009	1.6418	0.29684
30.483	3.2421e-009	1.6716	0.3034
30.583	5.0308e-009	1.6991	0.30995
30.683	3.9373e-009	1.725	0.31647
30.783	2.6661e-009	1.7496	0.32296
30.883	1.4842e-009	1.7725	0.32943
30.984	1.4007e-009	1.7967	0.33586
31.044	2.0411e-009	1.8104	0.33972
31.054	1.6582e-009	1.819	0.34052
31.064	4.3683e-009	1.8267	0.34131
31.094	3.6806e-009	1.8452	0.34365
31.184	5.3216e-009	1.887	0.35059
31.284	5.781e-009	1.9258	0.35824
31.384	5.6189e-009	1.9605	0.36585
31.484	2.056e-009	1.9927	0.37341
31.584	4.1665e-009	2.0235	0.38094
31.685	3.1292e-009	2.0523	0.38845
31.785	2.2823e-009	2.0796	0.39594
31.885	2.3226e-009	2.1059	0.40339
31.985	3.3369e-009	2.1317	0.41081
32.045	3.3883e-009	2.1468	0.41526
32.055	3.6251e-009	2.1535	0.41609
32.065	3.4109e-009	2.1594	0.41692
32.095	3.4657e-009	2.1741	0.4194
32.185	1.9794e-009	2.2099	0.42679
32.285	4.6938e-009	2.2447	0.43496
32.386	3.8686e-009	2.2766	0.4431
32.486	4.5854e-009	2.3064	0.45115
32.586	5.3083e-009	2.3349	0.45919
32.686	4.9109e-009	2.3622	0.46721
32.786	3.8124e-009	2.3883	0.47519
32.886	3.0696e-009	2.4137	0.48315
32.986	2.6197e-009	2.4382	0.49109
33.047	2.4486e-009	2.4525	0.49586
33.057	2.6866e-009	2.4515	0.49657
33.067	2.39e-009	2.4511	0.49728
33.097	1.3821e-009	2.4523	0.49944
33.187	1.2227e-009	2.4733	0.50594
33.287	1.2152e-009	2.4989	0.51316
33.387	3.774e-010	2.5228	0.52039
33.487	1.6533e-009	2.5458	0.52758
33.587	9.9191e-010	2.5681	0.53478
33.687	2.224e-009	2.5898	0.54198
33.788	1.9804e-009	2.611	0.54914
33.888	1.2967e-009	2.6319	0.55632
33.988	7.9016e-010	2.6523	0.56347
34.048	1.3045e-009	2.6645	0.56773
34.058	7.0183e-010	2.6664	0.56801
34.068	3.6501e-009	2.6684	0.56833
34.098	5.1987e-009	2.6736	0.56938
34.188	6.6592e-009	2.6797	0.57267
34.288	1.2994e-008	2.6782	0.57644
34.388	1.6938e-008	2.672	0.58027
34.489	1.9958e-008	2.6636	0.58421
34.589	6.052e-009	2.6544	0.58813
34.689	1.428e-008	2.645	0.59206
34.789	1.4724e-008	2.6494	0.59602
34.889	6.5007e-009	2.6772	0.60003
34.989	5.3004e-009	2.7029	0.604
35.049	5.5891e-009	2.7178	0.60638
35.059	4.9294e-009	2.7188	0.61543

35.069	2.2077e-008	3.0028	0.62372
35.095	4.4803e-008	3.575	0.64387
35.142	1.8801e-008	4.3062	0.67878
35.19	4.3131e-008	4.913	0.71359
35.239	8.2382e-008	5.4404	0.7482
35.288	1.0002e-007	5.9002	0.78258
35.337	9.6667e-008	6.3096	0.81671
35.387	7.783e-008	6.6795	0.85059
35.436	5.1936e-008	7.0175	0.88425
35.486	3.675e-008	7.3293	0.9177
35.536	5.4069e-008	7.6192	0.95096
35.586	7.0881e-008	7.8907	0.98404
35.636	1.1586e-007	8.1464	1.017
35.686	1.431e-007	8.4018	1.0497
35.736	1.6584e-007	8.6531	1.0824
35.786	1.8421e-007	8.8952	1.1149
35.836	1.9891e-007	9.1289	1.1473
35.886	2.0934e-007	9.355	1.1795
35.936	1.7873e-007	9.5742	1.2117
35.987	1.5578e-007	9.787	1.2438
36.019	1.4456e-007	9.92	1.2642
36.051	1.3588e-007	10.051	1.2846
36.061	3.2841e-007	11.716	1.3218
36.071	3.7142e-007	13.027	1.3561
36.082	3.8089e-007	14.171	1.3924
36.093	5.2076e-007	15.147	1.4281
36.104	5.1297e-007	16.013	1.4636
36.115	4.8234e-007	16.799	1.4991
36.126	4.3496e-007	17.525	1.5345
36.138	3.7612e-007	18.202	1.5698
36.149	3.1624e-007	18.839	1.6051
36.161	2.4122e-007	19.443	1.6404
36.172	1.7216e-007	20.017	1.6756
36.184	1.0656e-007	20.566	1.7107
36.195	5.2568e-008	21.092	1.7457
36.207	4.9501e-008	21.599	1.7807
36.218	9.2353e-008	22.087	1.8157
36.23	1.3766e-007	22.559	1.8505
36.241	1.7811e-007	23.016	1.8853
36.253	2.1232e-007	23.459	1.92
36.264	2.399e-007	23.889	1.9547
36.276	2.6081e-007	24.307	1.9892
36.288	2.752e-007	24.713	2.0238
36.299	2.8332e-007	25.11	2.0582
36.311	2.855e-007	25.496	2.0926
36.322	2.8212e-007	25.873	2.127
36.334	2.7357e-007	26.241	2.1613
36.346	2.6029e-007	26.6	2.1955
36.357	2.4271e-007	26.952	2.2297
36.369	2.2128e-007	27.296	2.2638
36.381	1.9641e-007	27.633	2.2979
36.393	1.6854e-007	27.964	2.3319
36.404	1.3807e-007	28.287	2.3659
36.416	1.054e-007	28.605	2.3998
36.428	7.095e-008	28.917	2.4337
36.439	3.5283e-008	29.223	2.4675
36.451	8.1938e-009	29.523	2.5013
36.463	4.1725e-008	29.819	2.535
36.474	8.015e-008	30.11	2.5687
36.486	1.1915e-007	30.395	2.6023
36.498	1.5832e-007	30.677	2.6359

36.51	1.9742e-007	30.954	2.6695
36.521	2.3624e-007	31.227	2.703
36.533	2.7462e-007	31.495	2.7365
36.545	3.1239e-007	31.76	2.7699
36.557	3.4943e-007	32.022	2.8033
36.568	3.8562e-007	32.279	2.8367
36.58	4.2088e-007	32.533	2.87
36.592	4.551e-007	32.784	2.9033
36.604	4.8824e-007	33.032	2.9366
36.615	5.2025e-007	33.277	2.9698
36.627	5.5108e-007	33.518	3.003
36.639	5.8072e-007	33.757	3.0361
36.651	6.0914e-007	33.993	3.0692
36.663	6.3636e-007	34.226	3.1023
36.674	6.6238e-007	34.457	3.1354
36.686	6.8721e-007	34.685	3.1684
36.698	7.1089e-007	34.91	3.2014
36.71	7.3344e-007	35.134	3.2343
36.721	7.5491e-007	35.355	3.2673
36.733	7.7534e-007	35.573	3.3002
36.745	8.0411e-007	35.79	3.333
36.757	8.1331e-007	36.004	3.3659
36.769	8.3096e-007	36.217	3.3987
36.78	8.4781e-007	36.427	3.4315
36.792	8.6392e-007	36.635	3.4642
36.804	8.7156e-007	36.842	3.497
36.816	8.368e-007	37.046	3.5297
36.828	8.0418e-007	37.249	3.5624
36.839	7.7369e-007	37.45	3.595
36.851	7.4529e-007	37.65	3.6277
36.863	7.1893e-007	37.847	3.6603
36.875	6.9458e-007	38.043	3.6929
36.887	6.8316e-007	38.238	3.7255
36.899	6.5164e-007	38.431	3.758
36.91	6.3292e-007	38.643	3.7905
36.922	6.1592e-007	38.859	3.823
36.934	6.0058e-007	39.074	3.8555
36.946	5.8678e-007	39.287	3.888
36.958	5.7444e-007	39.499	3.9205
36.969	5.6345e-007	39.71	3.9529
36.981	5.4286e-007	39.919	3.9853
36.993	4.6801e-007	40.127	4.0177
37.005	3.954e-007	40.334	4.0501
37.017	3.2537e-007	40.539	4.0824
37.029	2.5864e-007	40.743	4.1148
37.04	1.9685e-007	40.946	4.1471
37.052	1.4454e-007	41.146	4.1791
37.062	1.1746e-007	39.603	4.1696
37.07	2.981e-007	38.824	4.1641
37.079	2.1751e-007	38.148	4.1599
37.087	1.8103e-007	37.517	4.1562
37.096	1.5558e-007	36.942	4.153
37.105	3.8228e-007	36.419	4.15
37.114	5.0141e-008	35.942	4.1472
37.124	1.412e-007	35.5	4.1445
37.133	1.6559e-007	35.075	4.1419
37.143	1.7306e-007	34.665	4.1394
37.153	5.0468e-007	34.269	4.1369
37.164	6.9891e-007	33.884	4.1346
37.174	2.893e-007	33.541	4.1324
37.185	2.8241e-007	33.206	4.1302

37.196	2.62e-007	32.878	4.1281
37.207	3.1189e-007	32.555	4.1261
37.219	4.6243e-007	32.238	4.1242
37.231	4.1408e-007	31.926	4.1224
37.243	2.4931e-007	31.618	4.1206
37.255	7.5306e-007	31.315	4.1189
37.268	5.286e-007	31.202	4.1173
37.281	6.8813e-008	31.195	4.1158
37.295	5.7283e-007	31.178	4.1144
37.309	6.714e-007	31.15	4.1131
37.323	4.5339e-007	31.111	4.1119
37.338	2.4676e-007	31.063	4.1107
37.354	1.872e-007	31.006	4.1097
37.37	2.2823e-007	30.938	4.1088
37.386	4.7508e-007	30.861	4.1079
37.403	2.6612e-007	30.775	4.1072
37.421	3.894e-007	30.68	4.1066
37.439	9.4971e-008	30.575	4.1062
37.458	2.3628e-007	30.46	4.1058
37.478	5.3184e-007	30.337	4.1057
37.498	8.2261e-007	30.204	
37.52	9.7363e-007	30.062	4.1058
37.542	9.7557e-007	29.91	4.1062
37.565	9.9644e-007	29.748	4.1067
37.59	1.0225e-006	29.577	4.1074
37.615	6.7728e-007	29.395	4.1084
37.642	4.2803e-007	29.204	4.1097
37.671	6.2405e-007	29.001	4.1112
37.7	6.0613e-007	28.787	4.113
37.731	5.4885e-007	28.571	4.115
37.761	3.3335e-007	28.354	4.1173
37.792	1.6595e-007	28.137	4.1198
37.823	1.6394e-007	27.921	4.1224
37.854	2.7835e-007	27.704	4.1253
37.886	4.1129e-007	27.488	4.1283
37.918	2.8408e-007	27.273	4.1315
37.95	2.1314e-007	27.058	4.1349
37.982	2.4617e-007	26.845	4.1385
38.015	3.447e-007	26.632	4.1422
38.034	3.6175e-007	26.51	4.1444
38.053	3.4729e-007	26.389	4.1466
38.064	3.3049e-007	26.324	4.143
38.074	3.2655e-007	26.261	4.1398
38.088	2.2753e-007	26.17	4.1355
38.122	1.5607e-007	25.945	4.1262
38.156	3.0903e-007	25.705	4.1172
38.19	3.9506e-007	25.444	4.1083
38.226	4.5904e-007	25.165	4.0995
38.262	5.2203e-007	24.873	4.0907
38.299	5.6073e-007	24.571	4.0821
38.336	5.4179e-007	24.261	4.0735
38.374	5.0815e-007	23.946	4.0649
38.413	4.7415e-007	23.625	4.0564
38.452	4.4149e-007	23.3	4.0478
38.492	4.1117e-007	22.973	4.0392
38.533	3.8417e-007	22.642	4.0305
38.574	3.6083e-007	22.309	4.0218
38.617	1.3701e-007	21.975	4.013
38.66	3.2499e-007	21.638	4.0041
38.705	3.1171e-007	21.3	3.9951
38.751	3.0068e-007	20.96	3.9859

38.797	2.4043e-007	20.618	3.9768
38.845	1.671e-007	20.275	3.9704
38.895	1.0492e-007	20.009	3.9644
38.946	5.3748e-008	20.075	3.9578
38.998	1.8608e-008	20.156	3.9507
39.026	2.0993e-008	20.199	3.9468
39.055	3.4733e-008	20.239	3.9427
39.065	4.7748e-008	20.255	3.9362
39.075	2.9523e-007	20.265	3.9305
39.086	3.5162e-007	20.276	3.9244
39.118	3.0731e-007	20.313	3.9073
39.151	4.9408e-007	20.348	3.8891
39.184	3.8689e-007	20.378	3.8698
39.217	2.0666e-007	20.404	3.8495
39.25	9.6771e-008	20.426	3.8282
39.284	3.8575e-007	20.444	3.806
39.317	4.0035e-007	20.458	3.7829
39.351	3.8424e-007	20.469	3.759
39.385	3.758e-007	20.475	3.7343
39.419	3.7315e-007	20.478	3.7088
39.453	3.7487e-007	20.475	3.6828
39.487	9.0673e-008	20.467	3.6561
39.521	2.3676e-007	20.454	3.6288
39.555	1.7495e-007	20.435	3.6011
39.589	1.4301e-007	20.41	3.5732
39.623	1.4858e-007	20.379	3.546
39.658	1.8079e-007	20.342	3.5192
39.692	2.2379e-007	20.299	3.4933
39.726	2.6934e-007	20.249	3.4685
39.76	3.1424e-007	20.194	3.4449
39.795	3.5728e-007	20.133	3.4224
39.829	3.4432e-007	20.065	3.401
39.863	3.1427e-007	19.993	3.3803
39.898	2.8988e-007	19.914	3.3601
39.932	2.7058e-007	19.83	3.3405
39.967	2.5567e-007	19.741	3.3212
40.001	2.4437e-007	19.647	3.3025
40.036	2.3588e-007	19.548	3.2844
40.056	2.3185e-007	19.486	3.2738
40.066	1.4033e-007	19.456	3.2641
40.076	8.7885e-009	19.427	3.2551
40.09	1.3257e-007	19.385	3.2436
40.124	9.1975e-008	19.275	3.2177
40.159	3.0845e-008	19.158	3.1931
40.195	2.2993e-008	19.033	3.1694
40.232	4.9461e-008	18.901	3.1463
40.27	2.6542e-008	18.762	3.1238
40.309	3.1134e-008	18.615	3.1019
40.349	3.9602e-008	18.461	3.0803
40.389	8.2283e-008	18.299	3.0592
40.43	6.4312e-008	18.131	3.0384
40.472	5.5279e-008	17.955	3.018
40.515	4.1331e-008	17.774	2.9978
40.558	2.5442e-008	17.588	2.978
40.602	1.8507e-008	17.396	2.9585
40.647	2.3546e-008	17.199	2.9394
40.692	3.3207e-008	16.999	2.9208
40.738	4.2986e-008	16.794	2.9026
40.784	5.1775e-008	16.586	2.8848
40.831	5.9306e-008	16.451	2.8674
40.878	2.7483e-008	16.32	2.8504

40.926	4.467e-008	16.186	2.8339
40.974	2.7761e-008	16.048	2.8177
41.022	5.0147e-008	15.907	2.8023
41.058	6.8798e-008	15.802	2.7918
41.068	7.3629e-008	15.773	2.786
41.078	2.8394e-008	15.743	2.7807
41.1	7.5987e-008	15.676	2.7696
41.144	8.6612e-008	15.544	2.7497
41.189	8.8241e-008	15.406	2.73
41.236	9.3628e-008	15.261	2.7107
41.284	1.1736e-007	15.108	2.6915
41.334	1.2909e-007	14.949	2.6726
41.386	1.2252e-007	14.783	2.654
41.44	8.2207e-008	14.611	2.6356
41.495	7.3354e-008	14.432	2.6174
41.552	4.3303e-008	14.248	2.5994
41.61	1.3269e-007	14.058	2.5816
41.67	1.2047e-007	13.863	2.564
41.731	9.4326e-008	13.664	2.5466
41.793	3.7868e-008	13.461	2.5296
41.856	4.8227e-008	13.255	2.5128
41.921	5.0053e-008	13.046	2.4964
41.986	3.9689e-008	12.835	2.4803
42.023	2.459e-008	12.716	2.4715
42.059	1.1135e-008	12.598	2.4628
42.069	3.7333e-008	12.566	2.4587
42.079	6.8415e-008	12.534	2.4547
42.109	3.3786e-008	12.437	2.4433
42.17	1.0017e-007	12.242	2.421
42.234	1.293e-007	12.039	2.3982
42.301	1.2497e-007	11.827	2.375
42.371	1.0418e-007	11.607	2.3514
42.444	3.027e-008	11.379	2.3277
42.519	5.7887e-008	11.145	2.3038
42.598	9.9896e-008	10.905	2.2799
42.679	8.4192e-008	10.661	2.256
42.762	3.1823e-008	10.413	2.2322
42.847	3.6175e-008	10.163	2.2085
42.934	4.3028e-008	9.9107	2.1849
43.023	1.9628e-008	9.6582	2.1616
43.061	1.164e-008	9.5504	2.1517
43.071	3.6759e-008	9.5221	2.1479
43.081	3.8742e-008	9.4939	2.1442
43.111	1.6435e-008	9.4097	2.1335
43.143	5.4676e-008	9.3197	2.1223
43.228	6.054e-008	9.0861	2.0935
43.318	6.2194e-008	8.8427	2.0641
43.413	2.8186e-008	8.5905	2.0343
43.513	6.5837e-008	8.3311	2.0042
43.613	6.9181e-008	8.0766	1.9752
43.714	3.5984e-008	7.8277	1.9473
43.814	2.9796e-008	7.5844	1.9205
43.914	3.3519e-008	7.3467	1.8947
44.014	1.4241e-008	7.1145	1.8698
44.062	6.8118e-009	7.0044	1.8582
44.072	2.5367e-008	6.9815	1.8549
44.082	2.1337e-008	6.9586	1.8517
44.112	2.1143e-008	6.8905	1.8425
44.172	3.5936e-008	6.7576	1.8246
44.272	4.3298e-008	6.5387	1.7957
44.372	4.2082e-008	6.3249	1.768

44.472	2.4556e-008	6.1648	1.7414
44.572	4.8781e-008	6.0824	1.716
44.672	2.3038e-008	6.0086	1.6917
44.772	2.5763e-008	5.9419	1.6685
44.872	2.676e-008	5.8818	1.6463
44.973	1.972e-008	5.829	1.6252
45.063	2.2551e-009	5.7849	1.6069
45.073	1.6424e-008	5.7613	1.6043
45.083	2.4256e-008	5.7404	1.6018
45.113	2.9694e-008	5.6885	1.5944
45.193	2.6471e-008	5.5819	1.5756
45.293	3.2179e-008	5.4727	1.553
45.393	2.6052e-008	5.3791	1.5313
45.493	2.2291e-008	5.2966	1.5107
45.594	2.0145e-008	5.2475	1.4909
45.694	1.6856e-008	5.2105	1.472
45.794	2.0852e-008	5.1722	1.4539
45.894	1.5378e-008	5.1327	1.4367
45.994	4.6451e-009	5.0925	1.4202
46.065	2.0488e-009	5.0646	1.409
46.075	1.0694e-008	5.0607	1.407
46.085	1.796e-008	5.0567	1.405
46.115	2.4517e-008	5.0447	1.3992
46.205	1.2187e-008	5.0077	1.3826
46.305	2.4744e-008	4.9656	1.3651
46.405	1.8137e-008	4.9225	1.3484
46.505	1.7197e-008	4.8785	1.3324
46.606	1.1337e-008	4.8336	1.3171
46.706	1.6938e-008	4.788	1.3025
46.806	1.6889e-008	4.7416	1.2883
46.906	8.9289e-009	4.6945	1.2746
47.006	1.0643e-009	4.6468	1.2614
47.066	3.1711e-009	4.618	1.2537
47.076	7.4907e-009	4.6132	1.252
47.086	1.3696e-008	4.6084	1.2505
47.116	1.805e-008	4.5939	1.2458
47.206	5.9006e-009	4.5499	1.2322
47.307	1.5047e-008	4.5006	1.2175
47.407	1.4989e-008	4.4509	1.2034
47.507	9.7708e-009	4.4007	1.1896
47.607	1.165e-008	4.3502	1.1763
47.707	1.3757e-008	4.2993	1.1634
47.807	1.2952e-008	4.262	1.1508
47.907	5.7902e-009	4.2317	1.1386
48.008	7.8262e-010	4.2032	1.1267
48.068	3.6096e-009	4.1867	1.1197
48.078	5.6329e-009	4.1839	1.1182
48.088	1.0686e-008	4.1811	1.1168
48.118	1.1538e-008	4.1726	1.1127
48.208	6.555e-009	4.1429	1.1006
48.308	8.2598e-009	4.1064	1.0876
48.408	8.5979e-009	4.0688	1.0749
48.508	7.0861e-009	4.0315	1.0626
48.608	1.2292e-008	3.9951	1.0505
48.708	9.9695e-009	3.9601	1.0388
48.809	9.5325e-009	3.9267	1.0275
48.909	3.9132e-009	3.8948	1.0164
49.009	1.2978e-009	3.8646	1.0056
49.069	3.5853e-009	3.847	0.99922
49.079	4.4945e-009	3.844	0.99795
49.089	8.5186e-009	3.8411	0.9967

49.119	8.7561e-009	3.8321	0.99301
49.209	7.3121e-009	3.8018	0.98222
49.309	5.1985e-009	3.7656	0.97057
49.409	5.7344e-009	3.7286	0.95923
49.51	5.8243e-009	3.6919	0.94819
49.61	1.1167e-008	3.6561	0.93743
49.71	7.7724e-009	3.6215	0.92695
49.81	7.3033e-009	3.5884	0.91675
49.91	2.8414e-009	3.5566	0.90682
50.01	1.4348e-009	3.5262	0.89716
50.07	3.2689e-009	3.5085	0.89148
50.08	3.7273e-009	3.5055	0.89036
50.09	6.9353e-009	3.5025	0.88926
50.12	7.2922e-009	3.4935	0.88601
50.211	6.8193e-009	3.4638	0.87651
50.311	3.9477e-009	3.4288	0.86623
50.411	4.4727e-009	3.3932	0.85623
50.511	4.2134e-009	3.358	0.84649
50.611	8.9072e-009	3.3236	0.837
50.711	6.347e-009	3.2903	0.82774
50.811	5.8953e-009	3.2582	0.81869
50.912	2.2669e-009	3.2273	0.80984
51.012	1.2901e-009	3.1977	0.80118
51.072	2.7795e-009	3.1804	0.79606
51.082	3.1526e-009	3.1774	0.79506
51.092	5.7608e-009	3.1745	0.79407
51.122	6.0219e-009	3.1657	0.79116
51.212	5.3787e-009	3.1372	0.78262
51.312	3.3759e-009	3.1039	0.77334
51.412	3.8126e-009	3.0702	0.76426
51.512	3.0978e-009	3.0368	0.75537
51.613	7.6449e-009	3.0042	0.74667
51.713	5.1391e-009	2.9726	0.73815
51.813	5.0383e-009	2.942	0.7298
51.913	2.0051e-009	2.9125	0.72163
52.013	9.7068e-010	2.8841	0.71362
52.073	2.1915e-009	2.8674	0.70888
52.083	2.7114e-009	2.8646	0.70796
52.093	4.8803e-009	2.8618	0.70706
52.123	4.8039e-009	2.8534	0.7044
52.213	3.3695e-009	2.8263	0.69657
52.314	3.6009e-009	2.7949	0.68806
52.414	3.9234e-009	2.7631	0.67974
52.514	2.3043e-009	2.7317	0.67159
52.614	5.4381e-009	2.701	0.6636
52.714	4.4355e-009	2.6712	0.65579
52.814	4.576e-009	2.6423	0.64813
52.914	1.9748e-009	2.6143	0.64063
53.015	5.6029e-010	2.5873	0.63328
53.075	1.5422e-009	2.5715	0.62893
53.085	4.4045e-009	2.5696	0.63021
53.095	1.1861e-009	2.5677	0.63131
53.125	4.4015e-009	2.5636	0.63423
53.215	1.5114e-008	2.6206	0.64223
53.315	3.8557e-008	2.8277	0.65072
53.415	3.3026e-008	2.9942	0.65896
53.515	2.4946e-008	3.1358	0.66704
53.615	2.4862e-008	3.2605	0.67503
53.716	3.3609e-008	3.3726	0.68295
53.816	4.4483e-008	3.4753	0.69085
53.916	5.4968e-008	3.5704	0.69874

54.016	4.0716e-008	3.6593	0.70663
54.076	2.6839e-008	3.7107	0.71139
54.086	2.3343e-008	3.7193	0.71218
54.096	1.9992e-008	3.7279	0.71298
54.126	1.1673e-008	3.7531	0.71537
54.216	2.3036e-008	3.8248	0.72255
54.316	1.418e-008	3.9005	0.73057
54.417	1.1057e-008	3.9728	0.73862
54.517	2.9267e-009	4.042	0.74672
54.617	2.9197e-009	4.1084	0.75487
54.717	5.9667e-009	4.1723	0.76307
54.817	7.6347e-009	4.2339	0.77132
54.917	8.3116e-009	4.2933	0.77962
55.017	3.2265e-009	4.3508	0.78797
55.078	3.5508e-009	4.3846	0.79301
55.088	4.2226e-009	4.3092	0.7918
55.098	1.0498e-008	4.2594	0.79077
55.128	1.5105e-008	4.2761	0.78808
55.215	9.8318e-009	4.2816	0.78117
55.315	3.9894e-008	4.2477	0.77383
55.415	1.2649e-008	4.1912	0.76694
55.515	4.6192e-008	4.1235	0.76041
55.615	3.471e-008	4.0509	0.75417
55.715	2.3838e-008	3.9771	0.74819
55.815	2.0738e-008	3.904	0.74244
55.916	2.6133e-008	3.8327	0.73687
56.016	1.1819e-008	3.7638	0.73149
56.079	1.3884e-008	3.7213	0.72818
56.089	7.2283e-009	3.7144	0.72716
56.099	1.9601e-008	3.7075	0.72619
56.129	1.503e-008	3.6868	0.72337
56.219	1.5531e-008	3.6162	0.71525
56.319	1.6109e-008	3.5312	0.70657
56.419	1.6972e-008	3.4745	0.6982
56.52	1.9498e-008	3.4842	0.6901
56.62	1.8815e-008	3.4903	0.68225
56.72	1.3092e-008	3.4928	0.67462
56.82	6.7353e-009	3.4921	0.66721
56.92	1.1313e-009	3.4898	0.66
57.02	4.3952e-009	3.4883	0.65299
57.08	6.9809e-009	3.4864	0.64887
57.09	7.2267e-009	3.486	0.64819
57.1	7.4615e-009	3.4857	0.6475
57.13	8.0942e-009	3.4843	0.64546
57.22	4.373e-009	3.4783	0.63945
57.321	2.3156e-009	3.4694	0.63295
57.421	3.2045e-009	3.4585	0.62661
57.521	3.929e-009	3.4459	0.62045
57.621	4.8463e-009	3.4316	0.61446
57.721	2.588e-009	3.4159	0.60865
57.821	2.6614e-009	3.3989	0.60299
57.921	3.8005e-009	3.3807	0.5975
58.022	1.4246e-009	3.3615	0.59217
58.082	7.3646e-010	3.3497	0.58905
58.092	9.1487e-010	3.3477	0.58854
58.102	1.1214e-009	3.3457	0.58802
58.132	1.7386e-009	3.3396	0.5865
58.222	3.8852e-010	3.3208	0.58201
58.322	7.4093e-010	3.2991	0.57717
58.422	5.9304e-010	3.2768	0.57249
58.522	6.7823e-010	3.254	0.56798

58.622	6.0927e-010	3.2307	0.56364
58.723	5.6016e-010	3.2071	0.55946
58.823	4.4608e-010	3.183	0.55544
58.923	2.6178e-010	3.1587	0.55158
59.023	1.3333e-010	3.1342	0.54789
59.083	7.7553e-011	3.1194	0.54576
59.093	2.7114e-010	3.1169	0.54553
59.103	3.0173e-010	3.1145	0.5453
59.133	1.3647e-010	3.1071	0.54459
59.223	6.789e-010	3.0848	0.54249
59.323	1.2306e-009	3.06	0.54029
59.424	1.1794e-009	3.0353	0.53824
59.524	1.1407e-009	3.0107	0.53632
59.624	1.5345e-009	2.9865	0.53453
59.724	2.0461e-009	2.9625	0.53285
59.824	2.445e-009	2.9389	0.53128
59.924	2.7097e-009	2.9157	0.52979
60.024	2.5859e-009	2.8928	0.52838
60.084	2.1668e-009	2.8793	0.52758
60.095	2.1544e-009	2.877	0.52745
60.105	2.1418e-009	2.8748	0.52733
60.135	2.0921e-009	2.868	0.52696
60.225	1.8701e-009	2.8482	0.52588
60.325	1.2508e-009	2.8265	0.52473
60.425	6.0805e-010	2.8051	0.52365
60.525	1.2511e-009	2.7842	0.52262
60.625	1.1995e-009	2.7636	0.52163
60.725	7.2101e-010	2.7433	0.5207
60.826	3.4613e-010	2.7234	0.5198
60.926	1.249e-010	2.7038	0.51894
61.026	1.7338e-010	2.6846	0.51811
61.086	2.3846e-010	2.6731	0.51762
61.096	5.2596e-010	2.6712	0.51752
61.106	2.7902e-010	2.6693	0.51742
61.136	1.9201e-010	2.6637	0.51713
61.226	5.4419e-010	2.6469	0.51629
61.326	1.9878e-010	2.6286	0.51538
61.426	3.3782e-010	2.6105	0.51448
61.527	5.4358e-010	2.5928	0.51359
61.627	3.6831e-010	2.5752	0.51272
61.727	6.8195e-010	2.5579	0.51186
61.827	3.8619e-010	2.5409	0.511
61.927	5.0604e-010	2.5241	0.51015
62.027	4.4436e-010	2.5075	0.5093
62.087	3.5169e-010	2.4976	0.5088
62.097	4.3623e-010	2.4959	0.50869
62.107	3.5886e-010	2.4943	0.50859
62.137	4.0006e-010	2.4894	0.50829
62.227	5.6082e-010	2.4748	0.5074
62.328	9.8892e-010	2.4588	0.50642
62.428	1.0341e-009	2.4431	0.50545
62.528	9.6364e-010	2.4275	0.50449
62.628	3.441e-010	2.4121	0.50353
62.728	8.0315e-010	2.4041	0.50258
62.828	5.963e-010	2.3975	0.50164
62.928	2.7562e-010	2.3908	0.50069
63.029	3.0018e-010	2.3841	0.49975
63.089	3.832e-010	2.3801	0.49919
63.099	4.9207e-010	2.3795	0.49908
63.109	6.5045e-010	2.3788	0.49897
63.139	5.9858e-010	2.3768	0.49864

63.229	4.1563e-010	2.3708	0.49767
63.329	6.3137e-010	2.3642	0.49659
63.429	7.2049e-010	2.3576	0.49553
63.529	2.3488e-010	2.3509	0.49447
63.629	5.7052e-010	2.3443	0.49341
63.73	7.2483e-010	2.3376	0.49236
63.83	6.9585e-010	2.331	0.49132
63.93	3.6275e-010	2.3243	0.49027
64.03	8.2841e-011	2.3177	0.48923
64.09	1.6189e-010	2.3137	0.48861
64.1	4.2277e-010	2.313	0.48849
64.11	5.4651e-010	2.3124	0.48837
64.14	5.4361e-010	2.3104	0.48802
64.23	6.5596e-010	2.3044	0.48697
64.33	7.5193e-010	2.2977	0.4858
64.431	1.0759e-009	2.291	0.48465
64.531	1.5336e-010	2.2843	0.4835
64.631	6.1946e-010	2.2776	0.48235
64.731	3.9247e-010	2.2709	0.48121
64.831	5.0559e-010	2.2642	0.48007
64.931	4.6027e-010	2.2574	0.47893
65.031	1.6513e-010	2.2507	0.4778
65.092	9.732e-012	2.2466	0.47712
65.102	4.2228e-010	2.2459	0.477
65.112	3.9713e-010	2.2453	0.47687
65.142	3.8013e-010	2.2432	0.47649
65.232	6.8322e-010	2.2371	0.47537
65.332	8.0331e-010	2.2303	0.47413
65.432	8.4439e-010	2.2235	0.4729
65.532	2.1253e-010	2.2167	0.47167
65.632	7.2563e-010	2.2099	0.47045
65.732	5.0651e-010	2.2031	0.46923
65.833	3.1043e-010	2.1962	0.46802
65.933	4.0042e-010	2.1893	0.4668
66.033	2.3353e-010	2.1825	0.46559
66.093	1.0185e-010	2.1783	0.46487
66.103	3.5899e-010	2.1776	0.46473
66.113	4.4303e-010	2.177	0.4646
66.143	1.7378e-010	2.1749	0.4642
66.233	6.6674e-010	2.1687	0.46303
66.333	8.4992e-010	2.1618	0.46173
66.433	6.6241e-010	2.1548	0.46044
66.534	2.4948e-010	2.1479	0.45915
66.634	3.7e-010	2.1409	0.45786
66.734	5.8144e-010	2.134	0.45658
66.834	2.4626e-010	2.127	0.4553
66.934	2.7224e-010	2.12	0.45402
67.034	2.5229e-010	2.1131	0.45274
67.094	1.4534e-010	2.1089	0.45197
67.104	1.8778e-010	2.1082	0.45183
67.114	2.6637e-010	2.1075	0.4517
67.144	3.6355e-010	2.1054	0.45129
67.235	4.9227e-010	2.0991	0.45007
67.335	6.1631e-010	2.0921	0.44873
67.435	4.9767e-010	2.085	0.44739
67.535	2.5522e-010	2.078	0.44605
67.635	3.6367e-010	2.071	0.44471
67.735	3.9436e-010	2.0639	0.44338
67.835	2.3108e-010	2.0569	0.44205
67.936	1.699e-010	2.0499	0.44072
68.036	1.4545e-010	2.0428	0.4394

68.096	1.5437e-010	2.0386	0.4386
68.106	2.1334e-010	2.0379	0.43846
68.116	9.8369e-011	2.0372	0.43832
68.146	3.7779e-010	2.0351	0.43791
68.236	3.3299e-010	2.0287	0.43666
68.336	4.3119e-010	2.0217	0.43529
68.436	3.337e-010	2.0146	0.43392
68.536	2.388e-010	2.0075	0.43255
68.637	2.5532e-010	2.0005	0.43119
68.737	2.0154e-010	1.9934	0.42983
68.837	2.4186e-010	1.9863	0.42847
68.937	7.2331e-011	1.9793	0.42711
69.037	1.1021e-010	1.9723	0.42576
69.097	9.9719e-011	1.968	0.42494
69.107	8.4329e-011	1.9673	0.4248
69.117	1.3419e-010	1.9666	0.42466
69.147	3.1019e-010	1.9645	0.42424
69.237	2.222e-010	1.9582	0.42299
69.338	2.9321e-010	1.9511	0.42161
69.438	2.1893e-010	1.9441	0.42022
69.538	2.0783e-010	1.937	0.41884
69.638	2.2157e-010	1.93	0.41746
69.738	1.3462e-010	1.923	0.41609
69.838	1.2099e-010	1.916	0.41472
69.938	1.8136e-011	1.909	0.41335
70.039	8.9044e-011	1.902	0.41198
70.099	7.2243e-011	1.8978	0.41117
70.109	4.3655e-011	1.8971	0.41103
70.119	1.5196e-010	1.8964	0.41089
70.149	2.2967e-010	1.8943	0.41047
70.239	1.4624e-010	1.888	0.40922
70.339	1.9548e-010	1.8811	0.40784
70.439	1.2672e-010	1.8741	0.40646
70.539	1.7039e-010	1.8671	0.40508
70.639	1.941e-010	1.8602	0.4037
70.739	1.1969e-010	1.8533	0.40233
70.84	6.093e-011	1.8464	0.40096
70.94	4.3682e-011	1.8395	0.3996
71.04	9.1877e-011	1.8326	0.39823
71.1	5.1207e-011	1.8285	0.39742

Material Data

Aluminum Alloy

TABLE 25
Aluminum Alloy > Constants

Density	2.77e-006 kg mm^-3
Coefficient of Thermal Expansion	2.3e-005 K^-1
Specific Heat	8.75e+005 mJ kg^-1 K^-1

TABLE 26
Aluminum Alloy > Color

Red	Green	Blue
138	104	46

TABLE 27
Aluminum Alloy > Compressive Ultimate Strength

Compressive Ultimate Strength MPa
0

TABLE 28
Aluminum Alloy > Compressive Yield Strength

Compressive Yield Strength MPa
280

TABLE 29
Aluminum Alloy > Tensile Yield Strength

Tensile Yield Strength MPa
280

TABLE 30
Aluminum Alloy > Tensile Ultimate Strength

Tensile Ultimate Strength MPa
310

TABLE 31
Aluminum Alloy > Isotropic Secant Coefficient of Thermal Expansion

Zero-Thermal-Strain Reference Temperature K
295.15

TABLE 32
Aluminum Alloy > Isotropic Thermal Conductivity

Thermal Conductivity W mm^-1 K^-1	Temperature K
0.114	173.15
0.144	273.15
0.165	373.15
0.175	473.15

TABLE 33
Aluminum Alloy > S-N Curve

Alternating Stress MPa	Cycles	R-Ratio
275.8	1700	-1
241.3	5000	-1
206.8	34000	-1
172.4	1.4e+005	-1
137.9	8.e+005	-1
117.2	2.4e+006	-1
89.63	5.5e+007	-1
82.74	1.e+008	-1
170.6	50000	-0.5
139.6	3.5e+005	-0.5
108.6	3.7e+006	-0.5
87.91	1.4e+007	-0.5
77.57	5.e+007	-0.5
72.39	1.e+008	-0.5
144.8	50000	0
120.7	1.9e+005	0
103.4	1.3e+006	0
93.08	4.4e+006	0
86.18	1.2e+007	0
72.39	1.e+008	0
74.12	3.e+005	0.5
70.67	1.5e+006	0.5
66.36	1.2e+007	0.5
62.05	1.e+008	0.5

TABLE 34
Aluminum Alloy > Isotropic Resistivity

Resistivity ohm mm	Temperature K
2.43e-005	273.15
2.67e-005	293.15
3.63e-005	373.15

TABLE 35
Aluminum Alloy > Isotropic Elasticity

Young's Modulus MPa	Poisson's Ratio	Bulk Modulus MPa	Shear Modulus MPa	Temperature K
71000	0.33	69608	26692	

TABLE 36
Aluminum Alloy > Isotropic Relative Permeability

Relative Permeability
1

TBC coating

TABLE 37
TBC coating > Constants

Density	5.9e-006 kg mm^-3
Specific Heat	4.6e+005 mJ kg^-1 K^-1
Thermal Conductivity	2.5e-003 W mm^-1 K^-1

TABLE 38
TBC coating > Color

Red	Green	Blue
155	244	255

TABLE 39
TBC coating > Melting Temperature

Melting Temperature K
2973.2

TABLE 40
TBC coating > Isotropic Elasticity

Young's Modulus MPa	Poisson's Ratio	Bulk Modulus MPa	Shear Modulus MPa	Temperature K
1.75e+005	0.275	1.2963e+005	68627	